



Anti-Inflammatory Drugs

- Part 2
- Dr. Shankar Chinta
- Katzung's Ch. 36



Glucocorticoids

NSAIDS

DMARDS

Anti-Gout

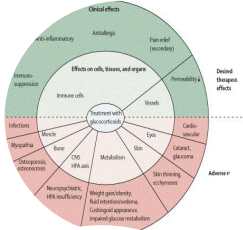
Anti-inflammatory Pharmacology



NSAIDs

(Non-steroidal anti-inflammatory drugs)

Role in blocking eicosanoids-mediated effects
Cost vs. Benefit of the three general classes
NSAID Choice



Glucocorticoids

(steroidal anti-inflammatory drugs)

Normal Function/intracellular mechanism
Anti-inflammatory effects
Types of synthetic glucocorticoids
Clinical pharmacology (uses and toxicity)



DMARDs

(Disease modifying anti-rheumatic drugs)

How do we stop disease progression with the least amount of harm?

Drugs for gout

Acute and chronic treatment



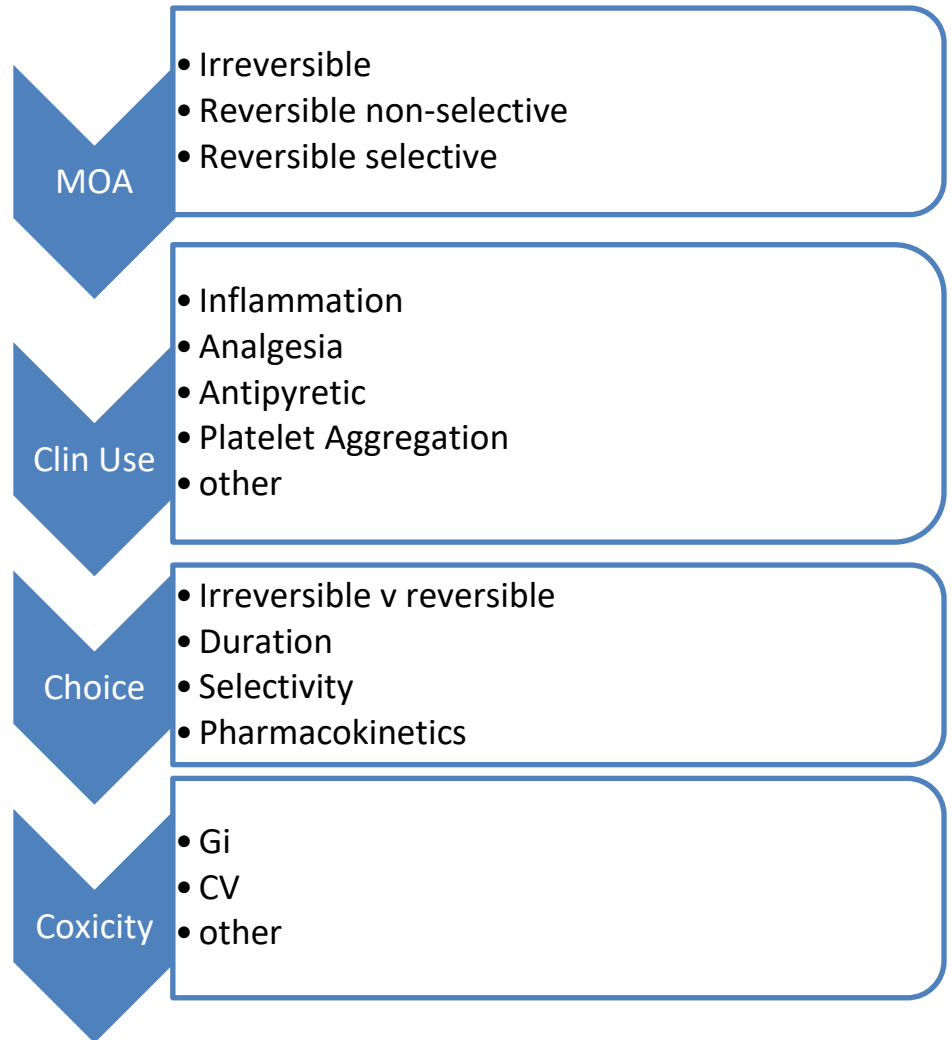
Glucocorticoids

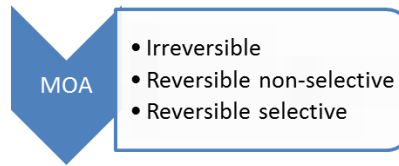
NSAIDS

DMARDs

Anti-Gout

NSAIDS





NSAID Classification

Aspirin

Traditional NSAIDS

Coxibs

Nonspecific
irreversible
inhibition of
cyclooxygenase 1
and 2 enzymes

Nonspecific
reversible inhibition
of cyclooxygenase 1
and 2 enzymes

Selective reversible
inhibition of
cyclooxygenase 2

NSAIDS

NSAIDS

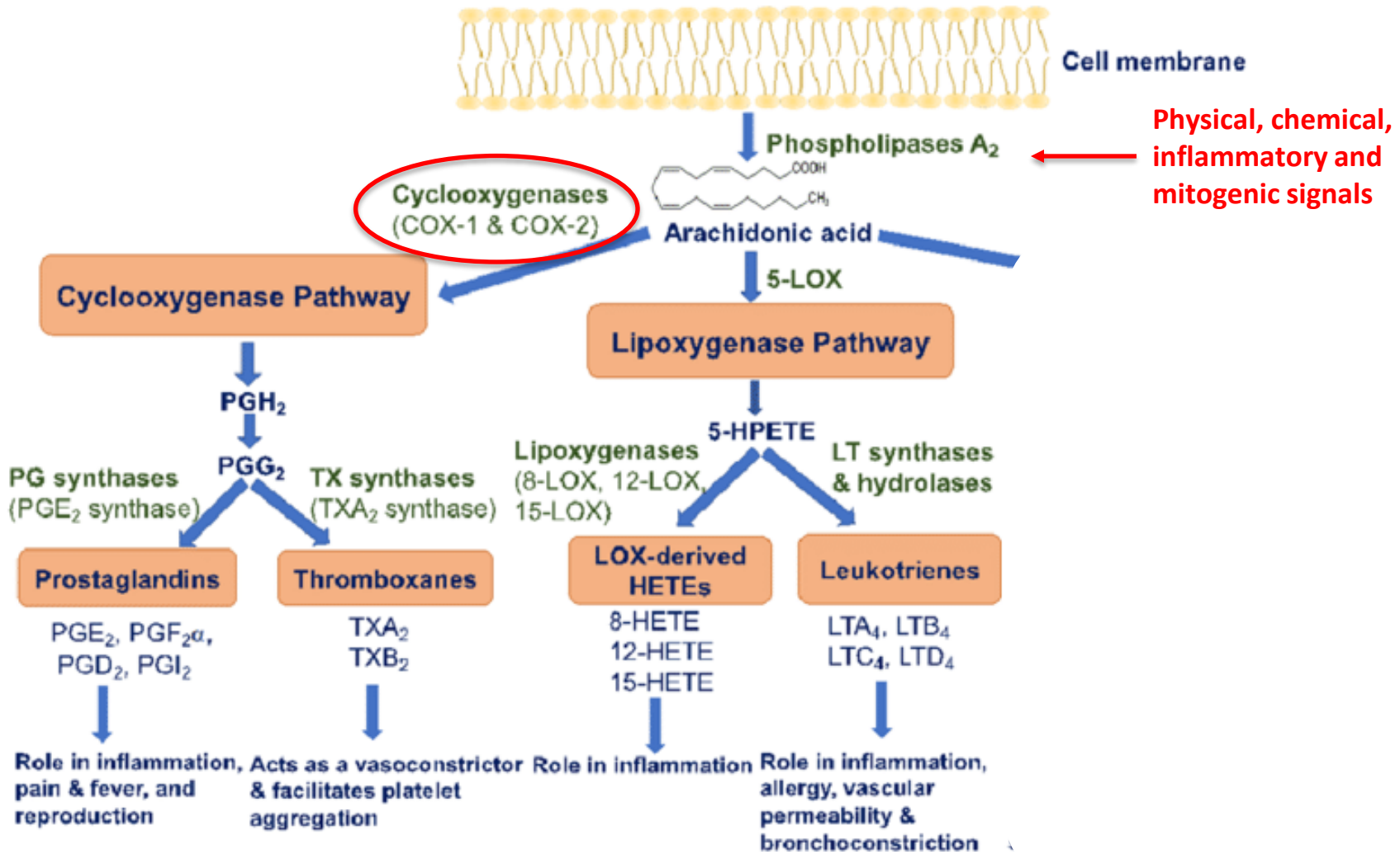


- Inflammation
- Analgesia
- Antipyretic
- Platelet Aggregation

Rx

- Inflammation
- Analgesia
- Antipyretic
- Platelet Aggregation

Synthesis of Arachidonic Acid Metabolites



Clin Use

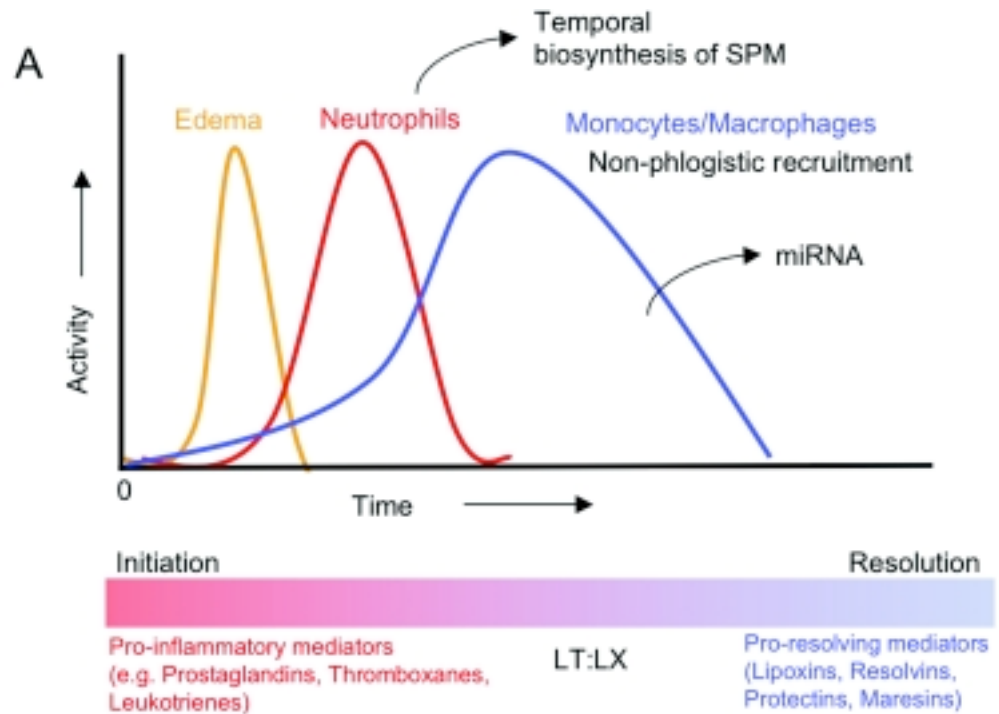
• **Inflammation**

- Analgesia
- Antipyretic
- Platelet Aggregation

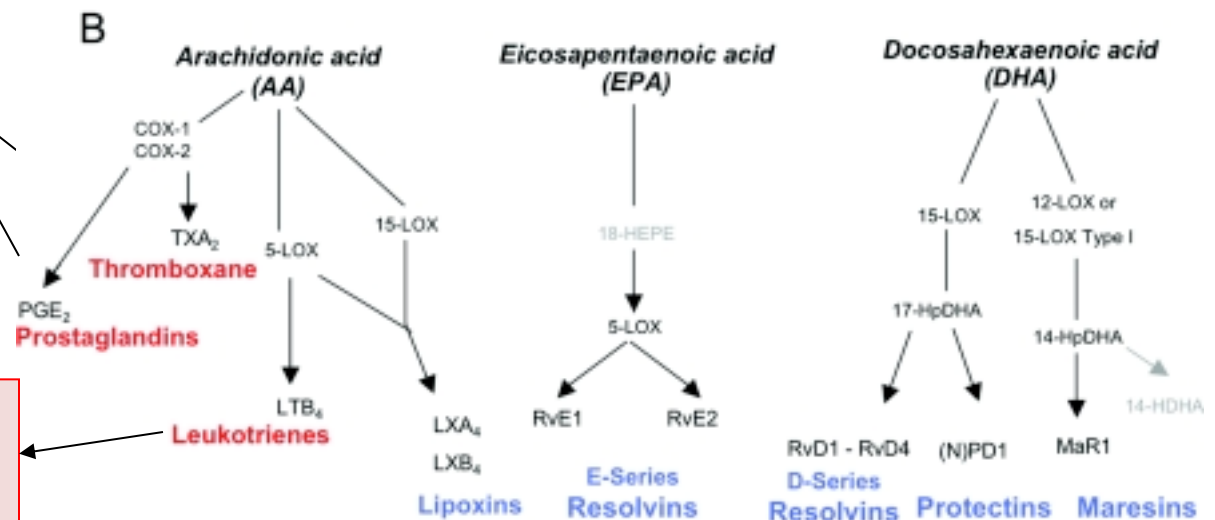
Immune cell chemo-attraction, activation, T cell modulation

éCapillary fluid outflow
Vascular permeability
↑endothelial cells gaps = swelling
ésecretion

Phagocyte attraction, activation, T cell modulation



Biochem. J. (2011) 437, 185–197



- **Inflammation**
- Analgesia
- Antipyretic
- Platelet Aggregation

Anti-inflammatory

effects of aspirin and NSAIDs

Rx

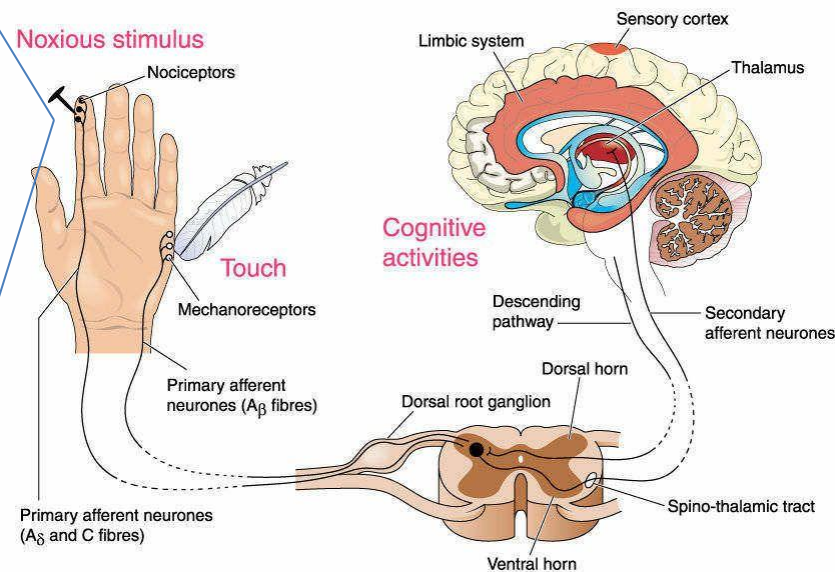
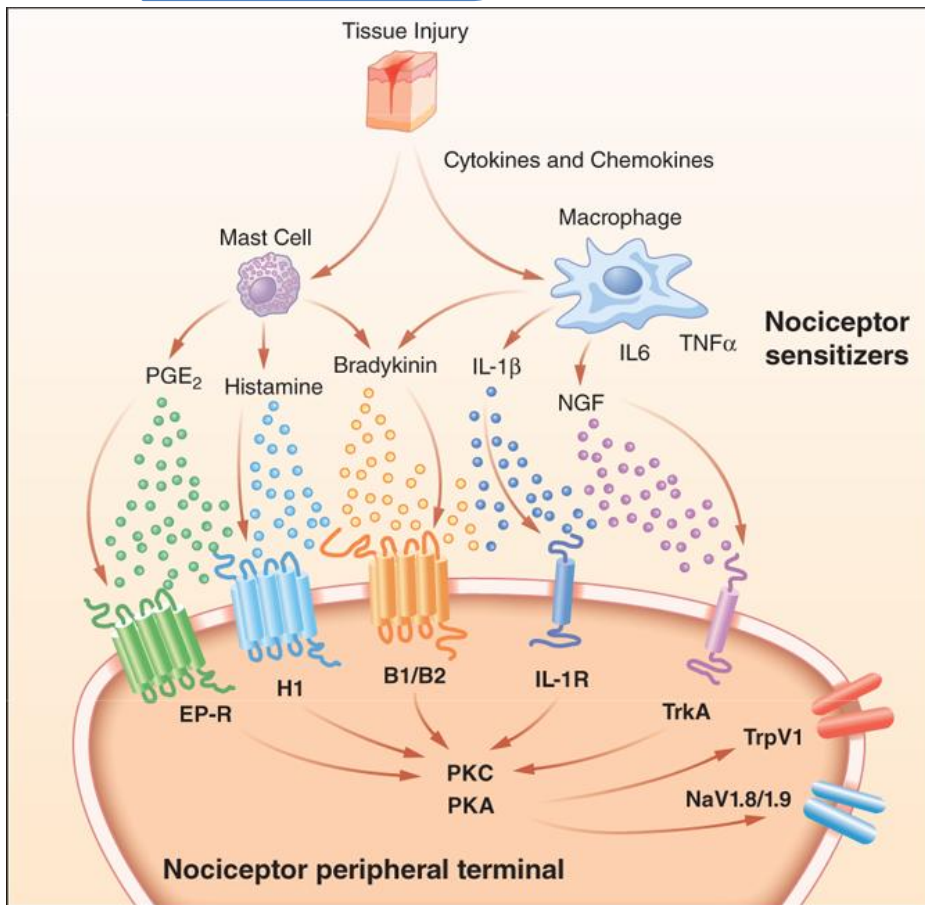
- Contribution to inflammation
 - Cox2>Cox 1
 - Cox1 early Cox2 later
- **Palliative treatment** for inflammatory joint conditions
 - rheumatoid arthritis
 - rheumatic fever
 - Osteoarthritis
 - Gout
 - Psoriatic arthritis
 - Arthritis associated with inflammatory bowel disease
- inflammation associated with mild bone and muscle trauma



Clin Use

- Inflammation
- Analgesia
- Antipyretic
- Platelet Aggregation

Eicosanoids Amplify Nociceptive Signals



Source: D.E. Longnecker, S.C. Mackey, M.F. Newman, W.S. Sandberg, W.M. Zapol: Anesthesiology, Third Edition Copyright © McGraw-Hill Education. All rights reserved.

NSAIDS

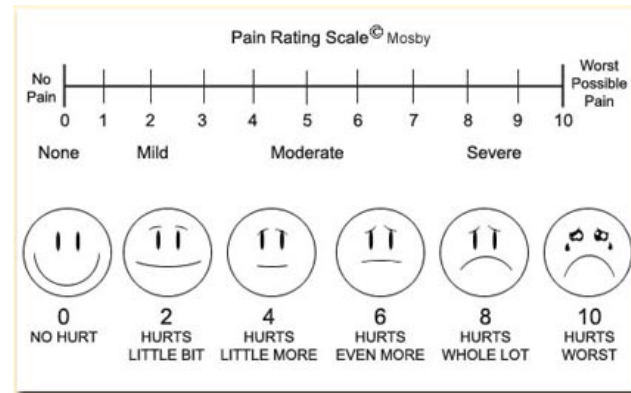
- Inflammation
- Analgesia
- Antipyretic
- Platelet Aggregation

Pain Management

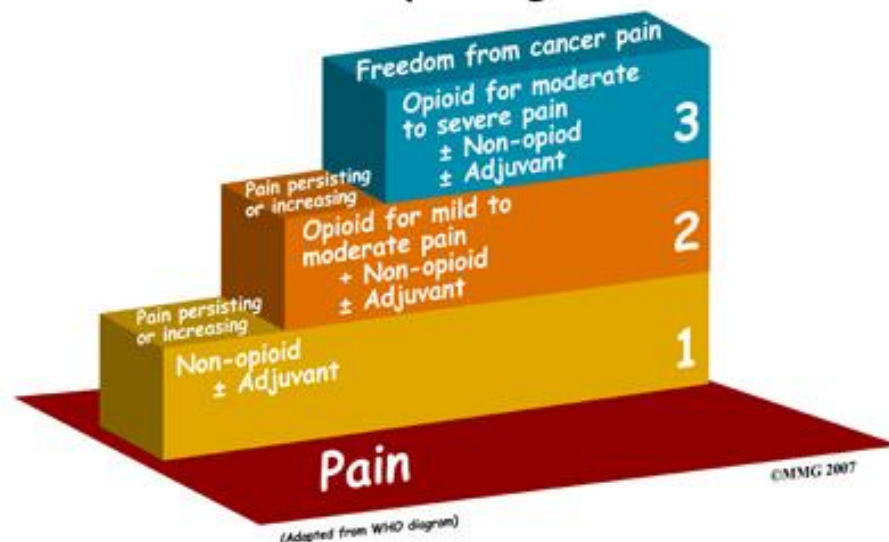


Analgesic Effects of Aspirin & Newer NSAIDS

- most effective on mild or moderate pain
 - e.g. Muscular, vascular, arthritis, bursitis, pain from dental origin and postpartum states
- Mechanism for pain inhibition
 - reduction of inflammation
 - inhibition of pain stimuli in neurc
- combined with other analgesics
 - no combination has been shown be more effective or less toxic th aspirin alone



WHO Three-Step Analgesic Ladder



Which NSAID should you use?

Always an issue and choice can be influenced by insurance formularies

Cost

Irreversible

Not necessary unless CV disease

Choice

Toxicity/
Benefit

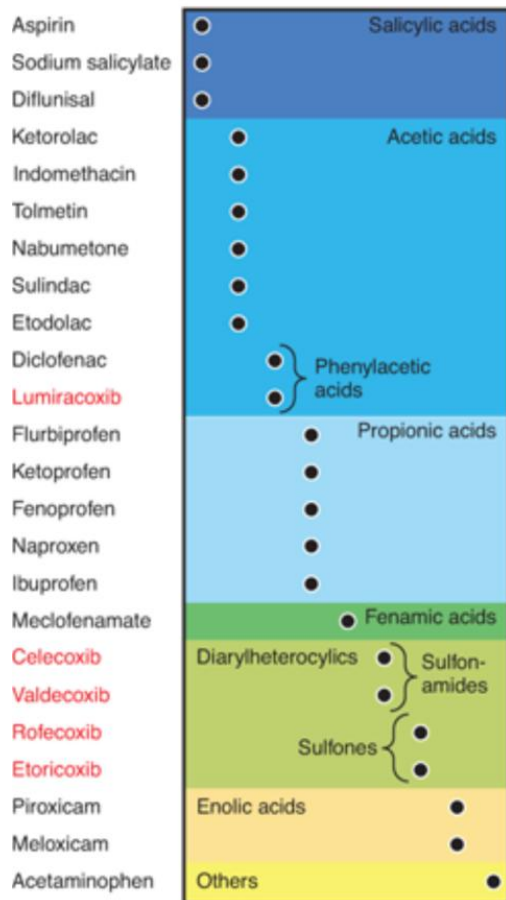
Disease
symptoms

Pain and
Inflammation
chronic
condition

Duration

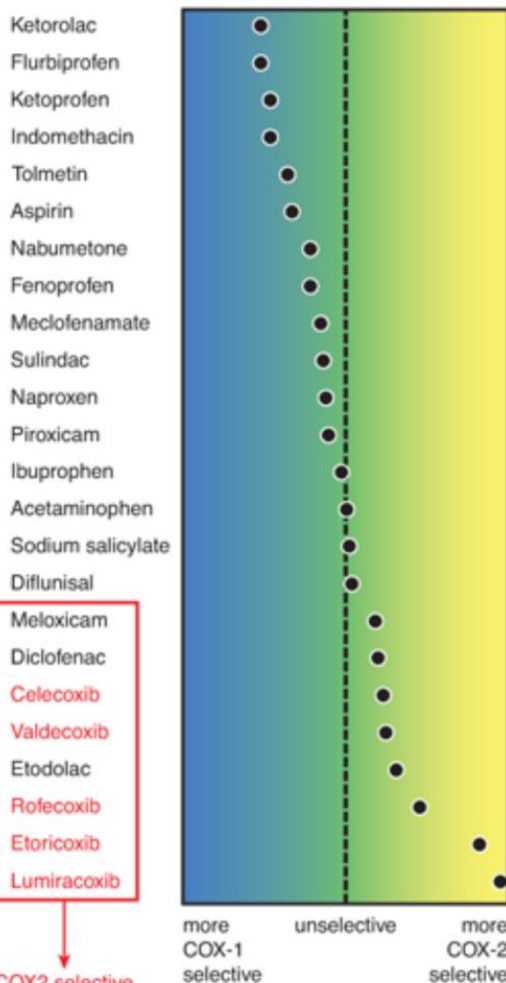
Minor consideration

Major consideration:
Sulindac (less renal toxicity).
Naproxen with CV disease,
Celecoxib better for GI? and
when patients on aspirin to
reduce clotting

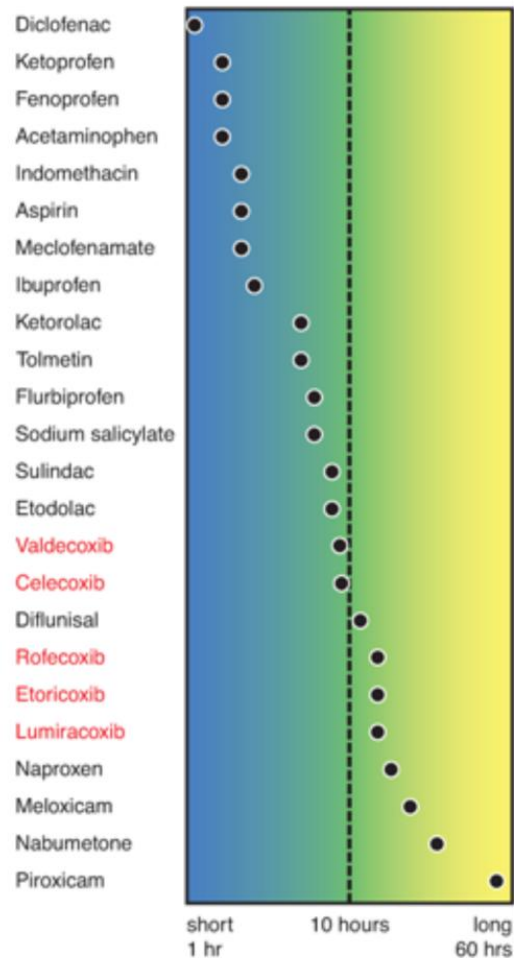
A Chemical similarity (arbitrary scale)

Black: tNSAID

Red: COX-2 inhibitors

B COX isoform selectivity (log scale)

COX2 selective NSAIDs

C Plasma $t_{1/2}$ (log scale)

short 1 hr 10 hours long 60 hrs

Source: L. L. Brunton, B. A. Chabner, B. C. Knollmann: Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 12ed.

www.accesspharmacy.com

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Classification of NSAIDs by chemical similarity(panel A), cyclooxygenase (COX) isoform selectivity (panel B), and plasma $t_{1/2}$ (panel C). The COX selectivity chart is plotted from data published in Warner et al., 1999, and FitzGerald and Patrono, 2001. tNSAIDs, traditional nonsteroidal anti-inflammatory drugs.

Source: Anti-inflammatory, Antipyretic, and Analgesic Agents; Pharmacotherapy of Gout, Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 12e

Citation: Brunton LL, Chabner BA, Knollmann BC. Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 12e; 2011 Available at:

<http://accessmedicine.mhmedical.com/content.aspx?bookid=1613§ionid=102161048> Accessed: October 26, 2017

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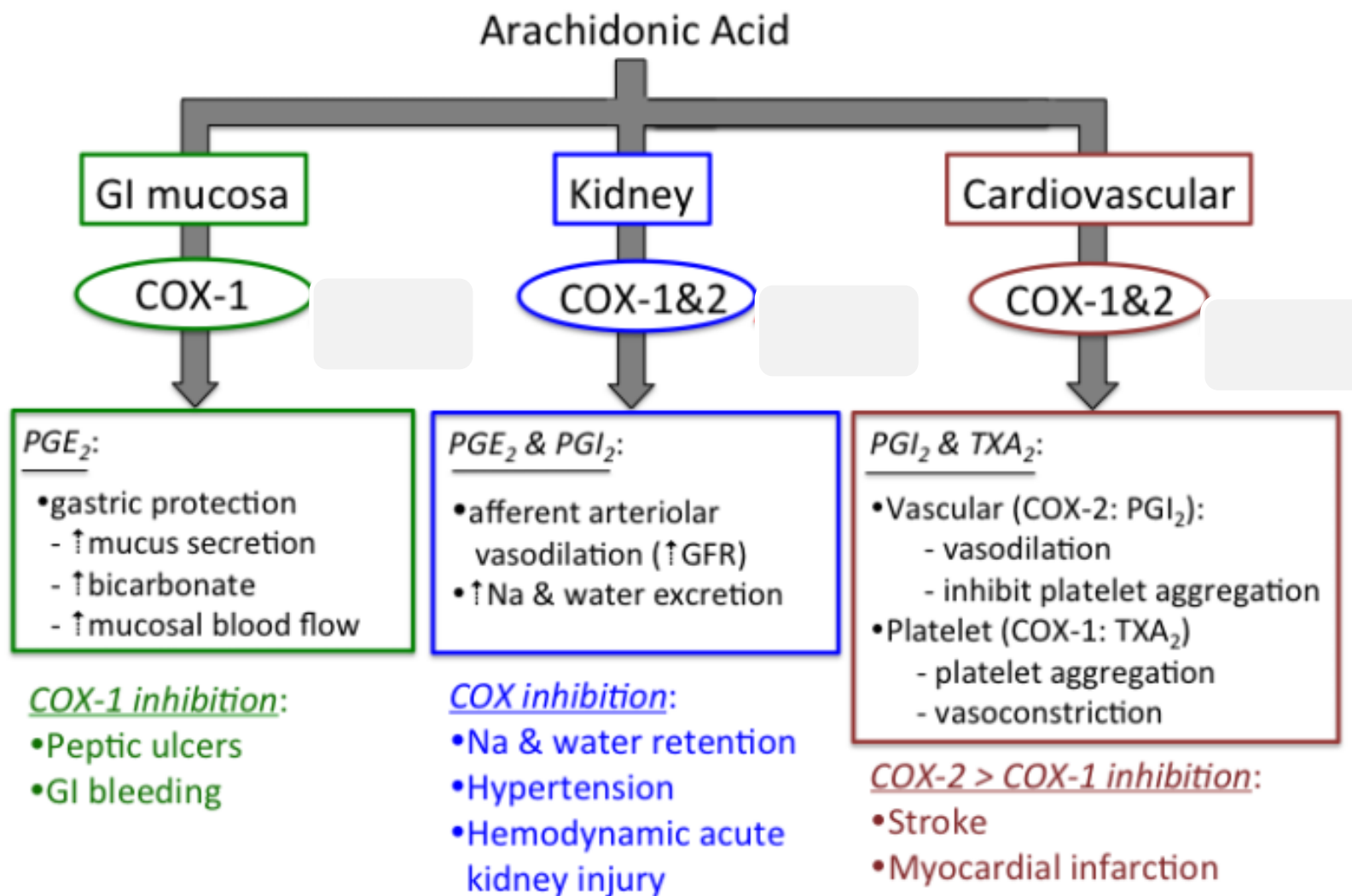
- Gi
- CV
- other

Summary of Adverse Events

Organ	Adverse Events
CNS	Tinnitus and dizziness
CV	Fluid retention, hypertension, edema, CHF
GI	Abdominal pain, ulcers and bleeding (significant cause of death in RA)
Hematologic	Thrombocytopenia , neutropenia, aplastic anemia, decreased platelet aggregation (Cox 1 v Cox 2)
Hepatic	Elevated liver enzymes, may cause mild asymptomatic hepatitis (10%), Especially with Diclofenac
Pulmonary	Asthma
Rashes	pruritus
Renal	Renal insufficiency, renal failure, hyperkalemia and proteinuria (much less with Sulindac)

Aspirin is associated with Reyes Syndrome

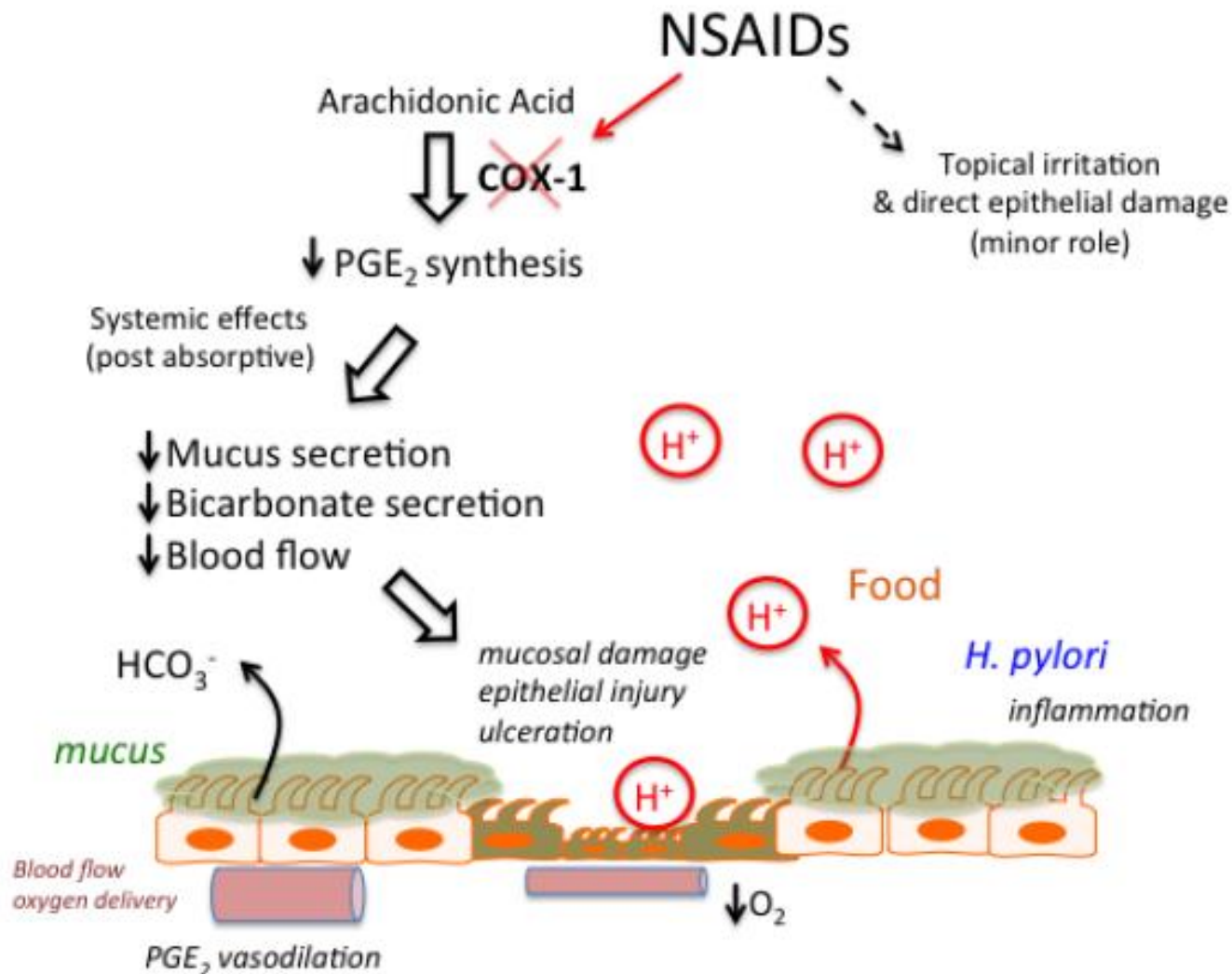
Three Major Side Effects Associated with NSAID Use



♦ Low dose aspirin irreversibly inhibits platelet COX-1

- Gi
- CV
- other

Gastrointestinal Effects

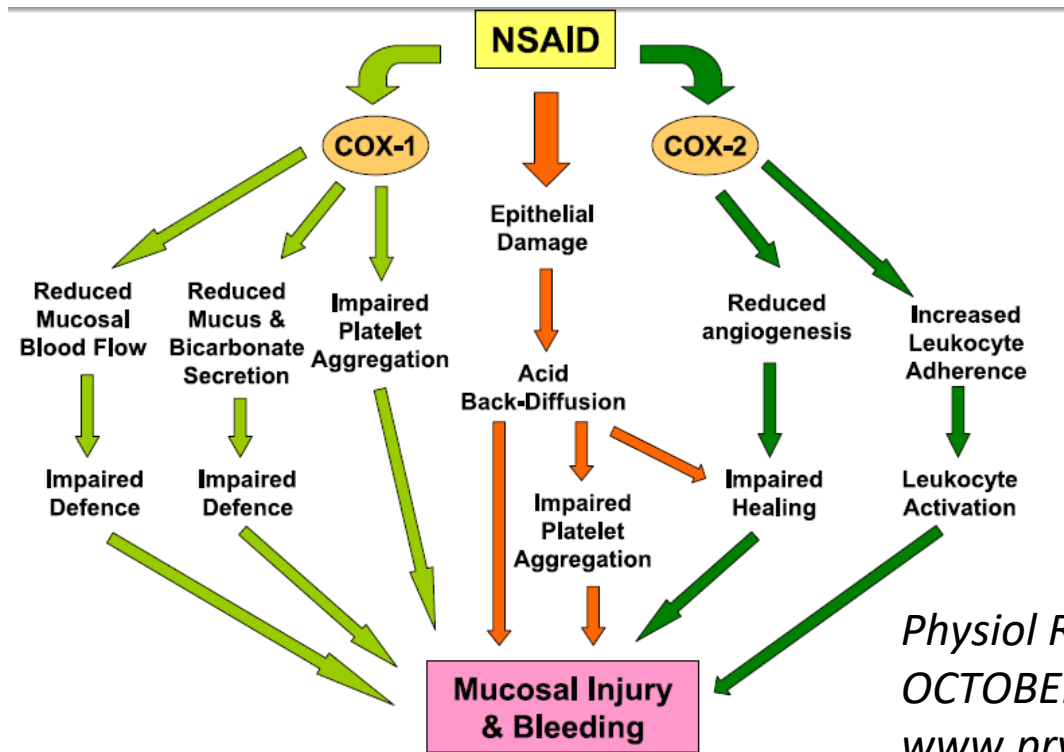


- Gi
- CV
- other

Gastrointestinal Effects

- Gastroduodenal ulceration & bleeding

COX-1 products are more involved in mucosal integrity and increase platelet aggregation



COX-2 are less involved but play an important role in angiogenesis and wound healing

Physiol Rev • VOL 88 •
OCTOBER 2008 •
www.prv.org

- Gi
- CV
- other

Gastrointestinal Effects

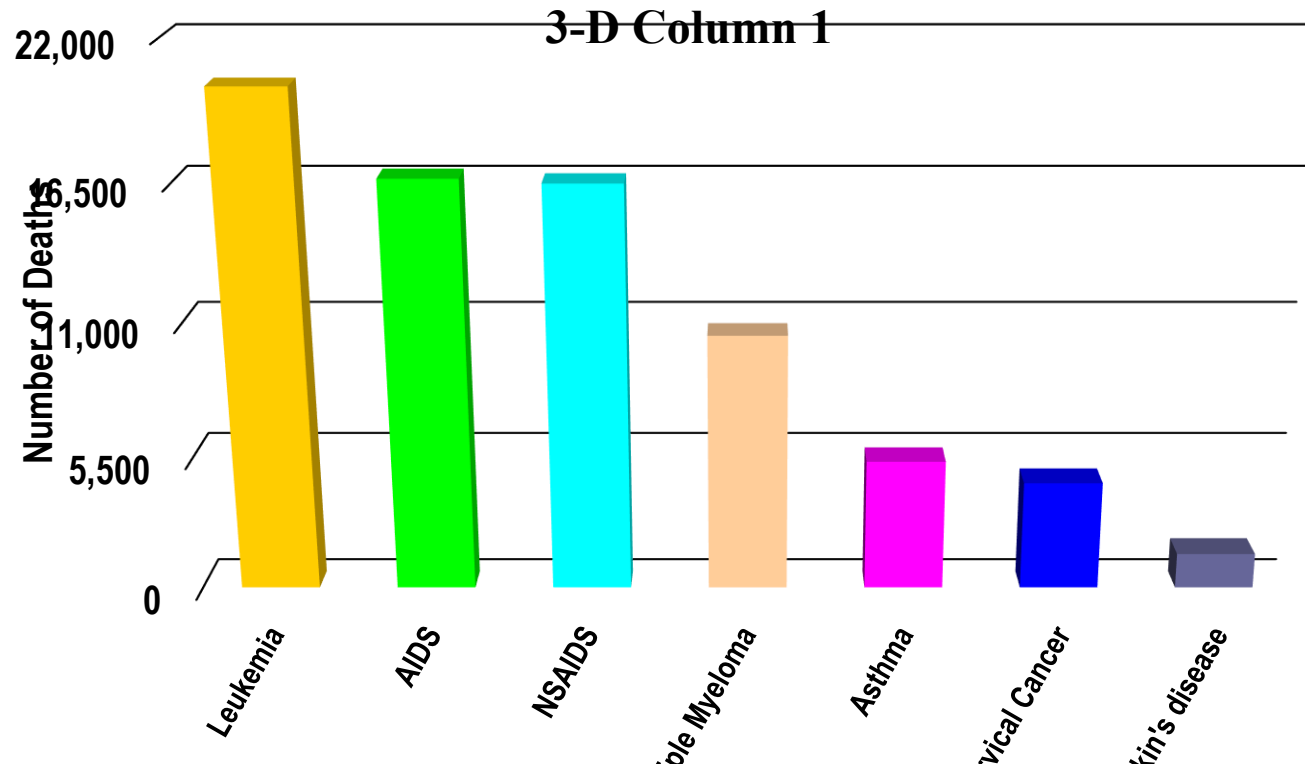
(Major problem with all NSAIDS)

- Gastric upset (intolerance, abdominal pain, severe nausea and/or dyspepsia)
- 10-20%
 - produced by:
 - undissolved tablets
 - ↓ of cytoprotective prostaglandins
 - minimize by:
 - taking w/ meals & antacids
 - enteric coated tablets
 - Proton pump inhibitors (omeprazole *Prilosec*)
 - Reduced with COX 2 specific inhibitors?
 - Note: buffered (not effective)

- Gi
- CV
- other

Gastrointestinal Effects

- Gastroduodenal ulceration & bleeding
 - 1-4% of chronic NSAID users (much higher in elderly)
 - Estimates of 16,500 NSAID induced deaths per year
 - Reduce with concomitant use of proton pump inhibitors or misoprostol or **with newer selective COX 2 inhibitors (\$\$\$\$\$\$)**



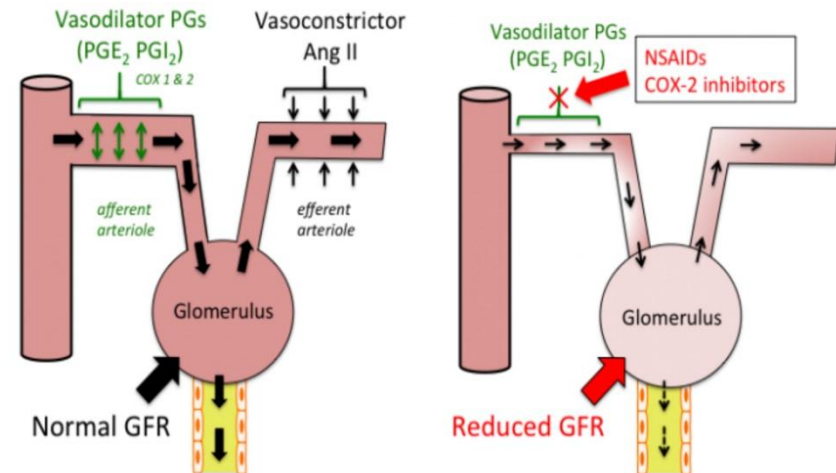
US mortality
Data (1997)
NEJM340:24

- Gi
- CV
- other

Adverse Effect's of

Aspirin and other NSAIDS Cont'd

- Kidneys:
 - Prostaglandins synthesized by both **COX1 and 2** promote renal vascular dilation and medullary blood flow
 - Blocking production of prostaglandins ⑨ unopposed vasoconstriction ⑨ renal ischemia
 - Not an issue in well hydrated patients with normal renal function
 - **Can be an issue in patients who have CHF, the elderly, and hypotensive patients or in patients with renal failure**
 - ↓ glomerular filtration rate
 - Acute renal failure (2/100,000)
 - Interstitial Nephritis



- Gi
- CV
- other

Cost/Benefit should factor into drug choice

Recommendations:

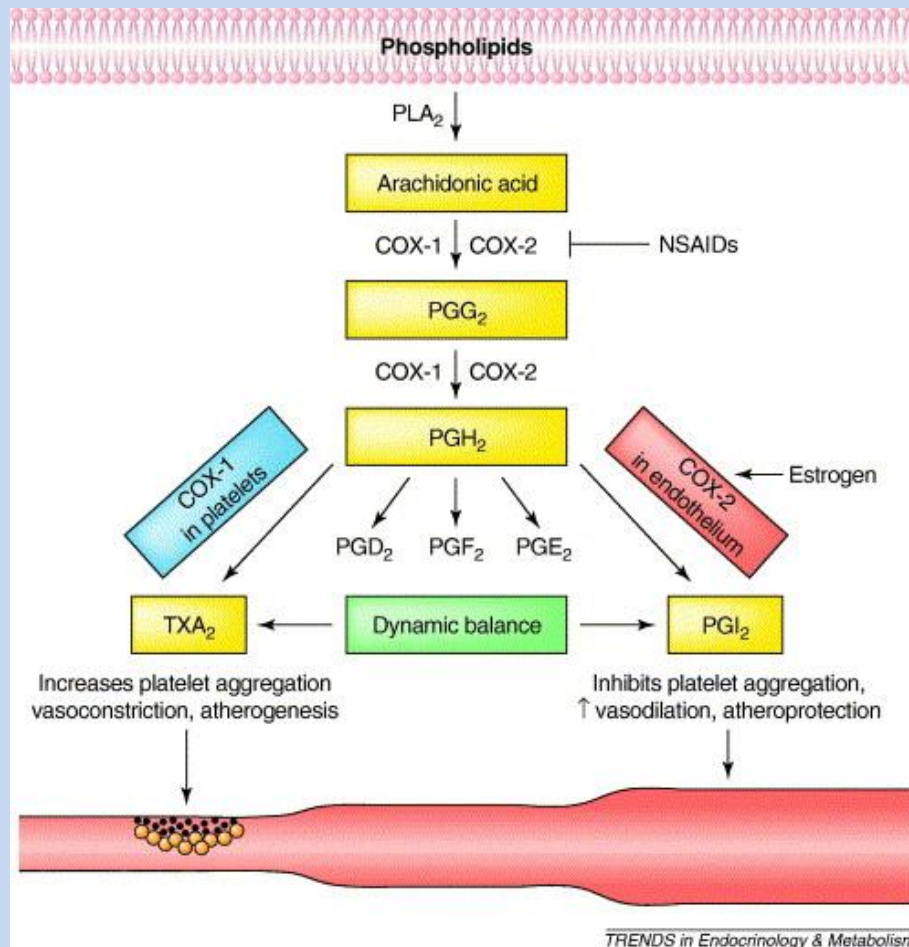
Coxibs may lead to less (50% ↓) GI toxicities than other NSAIDS if they are in the high risk group

- 72 + years (linear increase in risk)
- history of ulcers or GI bleeding
- Concomitant corticosteroid or anticoagulants use
- Smokers and alcoholics

Avoid use in patients with a history of cardiovascular issues

- Gi
- CV
- other

COX 2 Selective drugs are associated with ↑ risk of a CV event.



Most NSAIDs maintain balance

Inducible in endothelial cells, vascular protection during inflammation

COX 2

Selective inhibition shifts balance to coagulation

- Gi
- CV
- other

Drug Interactions: NSAIDS

- Low-dose aspirin combined with NSAID
 - Reversible prevents irreversible from binding
- + ACE inhibitors
 - Block prostaglandin-induced production of bradykinin and other vasodilators (covered in detail later)
 - Both may produce hyperkalemia in patients with renal or heart disease
- + Glucocorticoids
 - Both cause gastric irritation = ulcers
- + Warfarin = increased bleeding
 - NSAIDS decrease warfarin metabolism
 - NSAIDS inhibit platelet aggregation

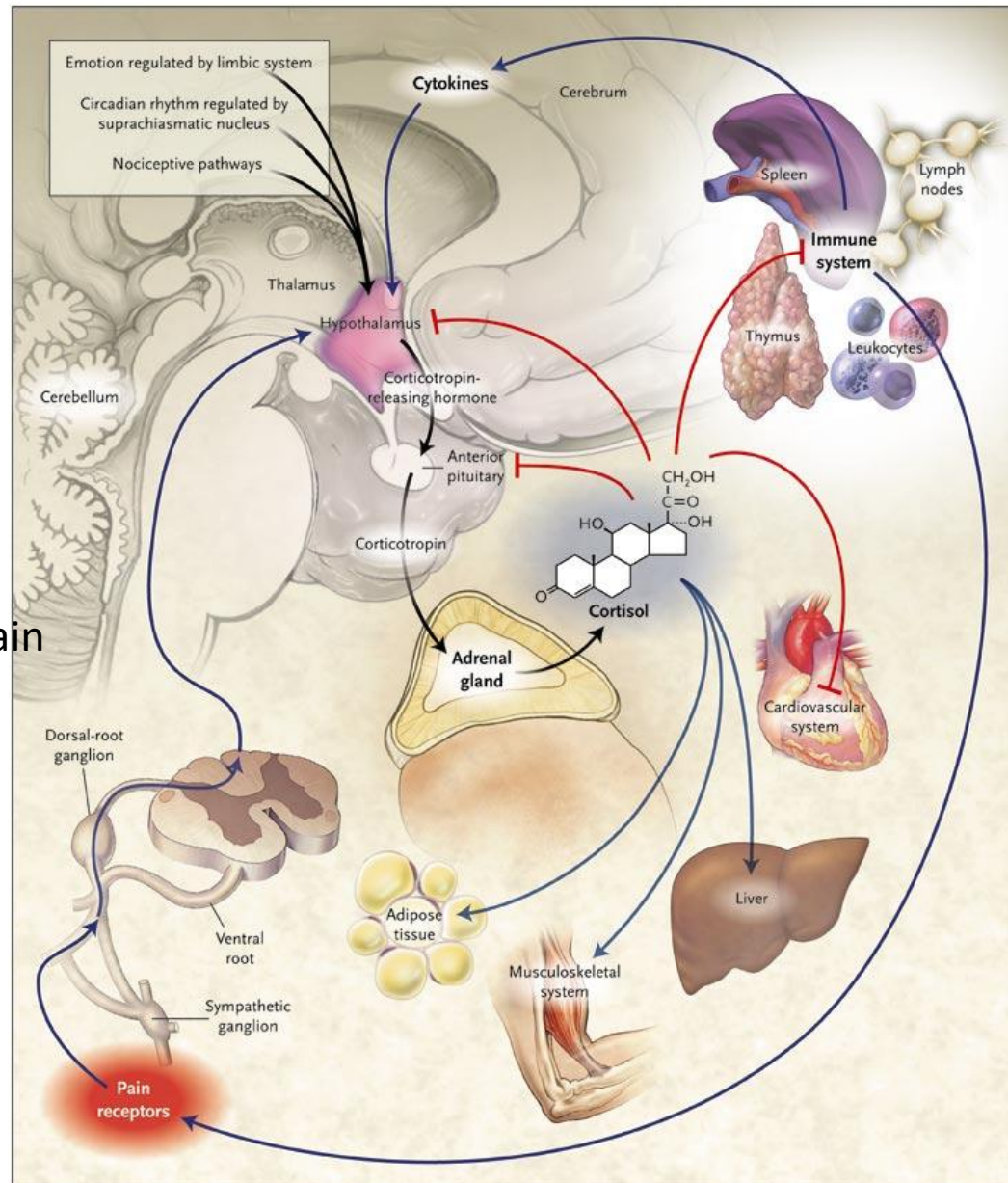
Glucocorticoids

Covered Today: Use to treat inflammation

Covered later: Use for adrenal disorders

Glucocorticoid Physiologic function

- Glucocorticoids; cortisol and corticosterone:
 - Glucocorticoids help regulate carbohydrate, protein and fat metabolism
 - ⑩ ↑ hepatic glucose output by stimulating the catabolism of peripheral fat and protein to provide substrate for hepatic gluconeogenesis
 - protects glucose-dependent brain and heart by increasing blood glucose levels
 - Endows organism with the ability to resist stressful stimuli
 - Modulate immune response (“dampens the immune-fire”)
 - Diminishes response to catecholamines



Glucocorticoid Anti-inflammatory Effects

↓ immune cells

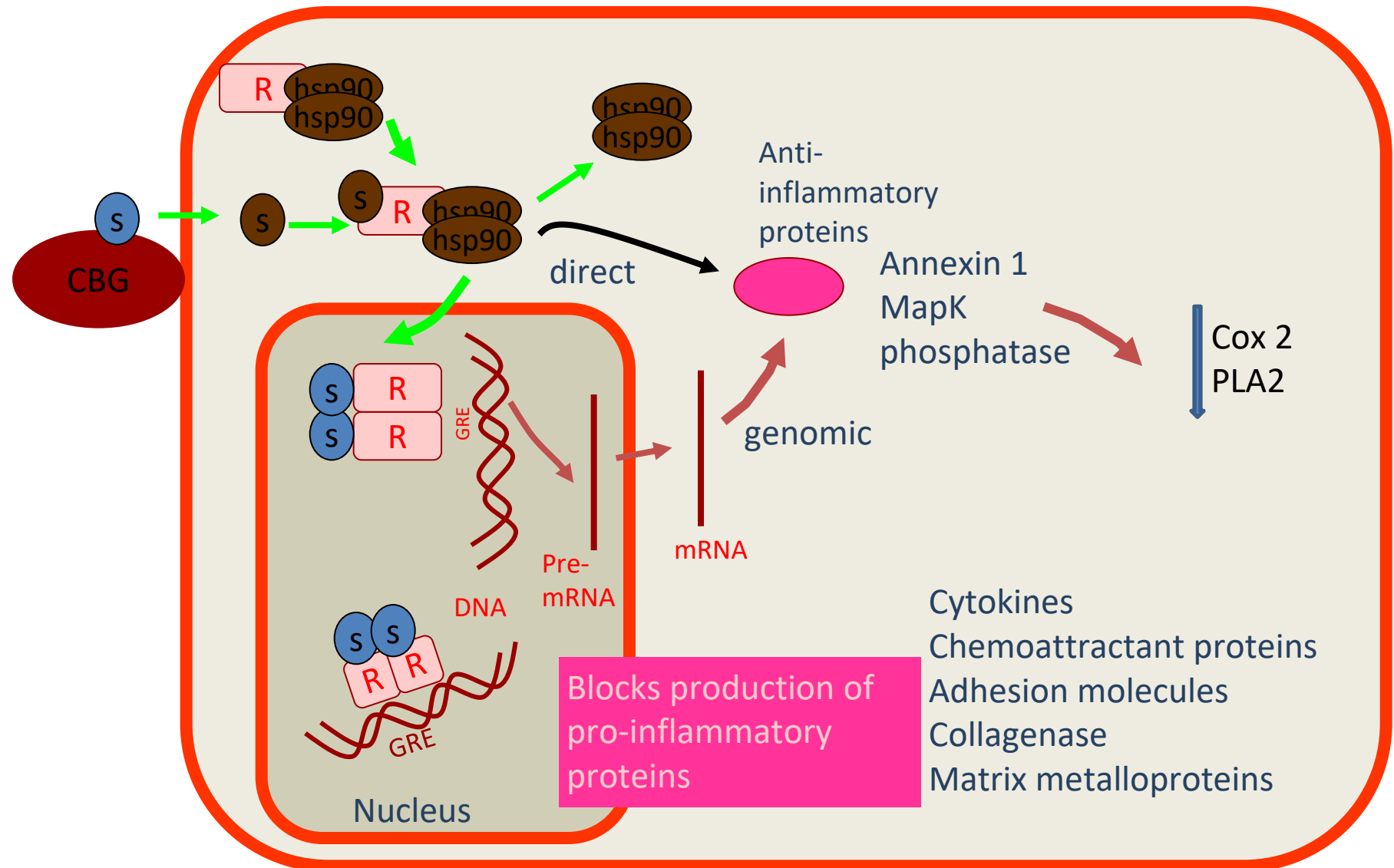
↓ arachidonic cascades

↓ # of neutrophils,
monocytes, lymphocytes,
basophils

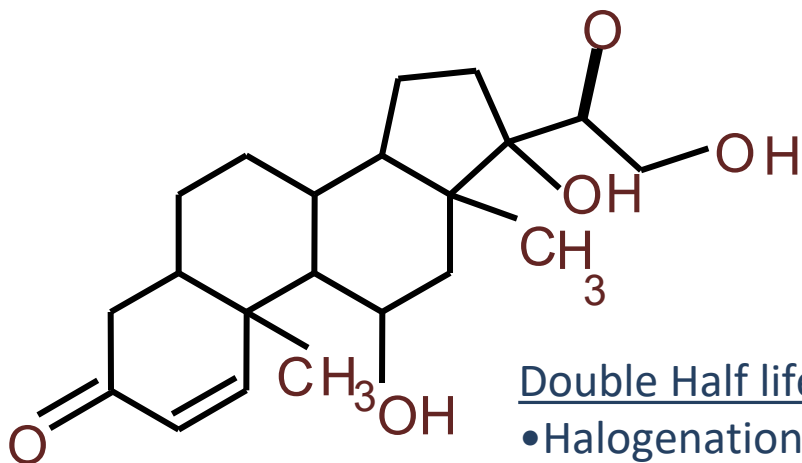
↓ cytokine production
↓ antibody production
↓ migration

↓ Phospholipase A2
↓ Cox 2 induction

Intracellular Mechanism of Glucocorticoid Action



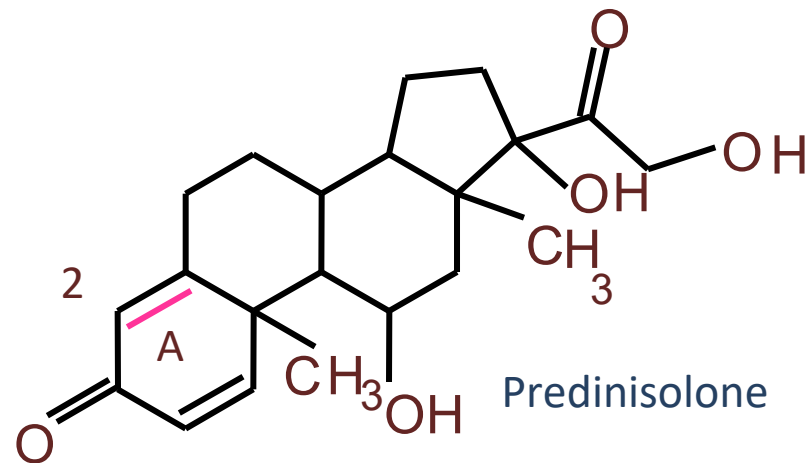
Synthetic Glucocorticoid Structures



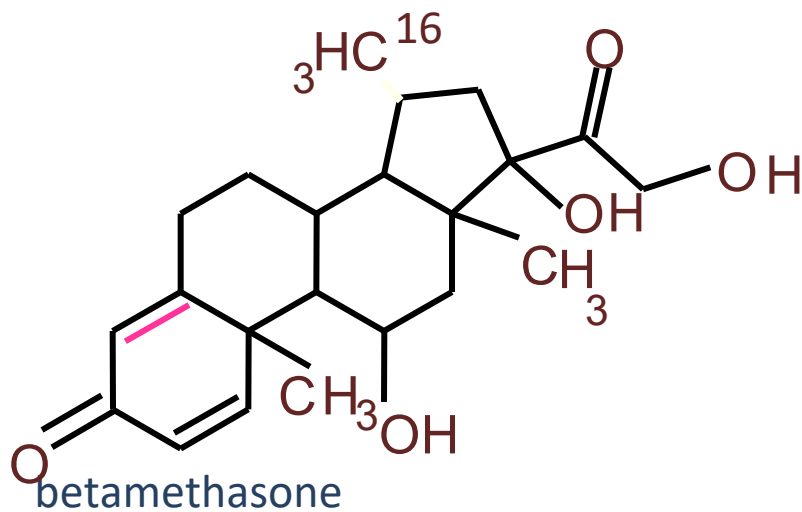
hydrocortisone

Double Half life:

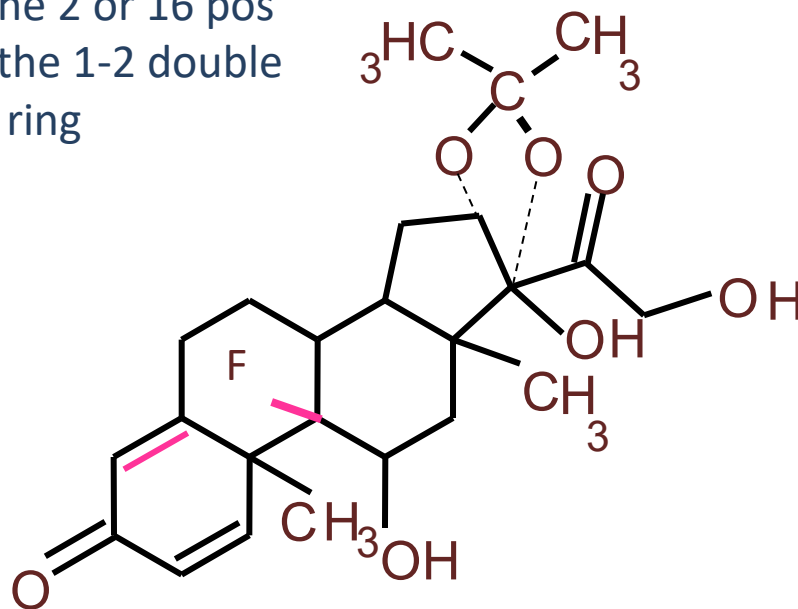
- Halogenation at the 9 pos.
- methylation at the 2 or 16 pos
- unsaturation of the 1-2 double bond on the A ring



Prednisolone



betamethasone



Triamcinolone (acetonide moiety shaded)

Synthetic Adrenocorticosteroids

Agent	Anti-inflam- matory	Activity ¹		Equivalent Oral Dose (mg)	Forms Available
		Topical	Salt- Retaining		
Short- to medium-acting glucocorticoids					
Hydrocortisone (cortisol)	1	1	1	20	Oral, injectable, topical
Cortisone	0.8	0	0.8	25	Oral, injectable, topical
Prednisone	4	0	0.3	5	Oral
Prednisolone	5	4	0.3	5	Oral, injectable, topical
Methylprednisolone	5	5	0	4	Oral, injectable, topical
Meprednisone ²	5		0	4	Oral, injectable
Intermediate-acting glucocorticoids					
Triamcinolone	5	5 ³	0	4	Oral, injectable, topical
Paramethasone ²	10		0	2	Oral, injectable
Fluprednisolone	15	7	0	1.5	Oral
Long-acting glucocorticoids					
Betamethasone	25–40	10	0	0.6	Oral, injectable, topical
Dexamethasone	30	10	0	0.75	Oral, injectable, topical
Mineralocorticoids					
Fludrocortisone	10	10	250	2	Oral, injectable, topical
Desoxycorticosterone acetate	0	0	20		Injectable, pellets

¹Potency relative to hydrocortisone.

²Outside USA.

³Acetonide: Up to 100.

Uses of glucocorticoids

Disorder	Examples
Allergic Reactions	dermitis, drug RX, rhinitis, angioedema, asthma, urticaria
Collagen vascular	Lupus erythematosus, rheumatoid arthritis
Eye diseases	Uveitis, conjunctivitis, optic neuritis
GI	Inflammatory bowel
Blood	leukemia, multiple myeloma, hemolytic anemia, thrombocytopenic purpura
Systemic inflammation	Acute respiratory distress syndrome
Bones and joints	arthritis, bursitis
Neurologic	cerebral edema, multiple sclerosis
Organ transplant	Prevent rejection
Pulmonary	asthma, infant respiratory distress syndrome, sarcoidosis
Renal disorders	Nephrotic syndrome
Thyroid	Thyroiditis
Miscellaneous	Hypercalcemia, mountain sickness

Table 39-2 Katzung

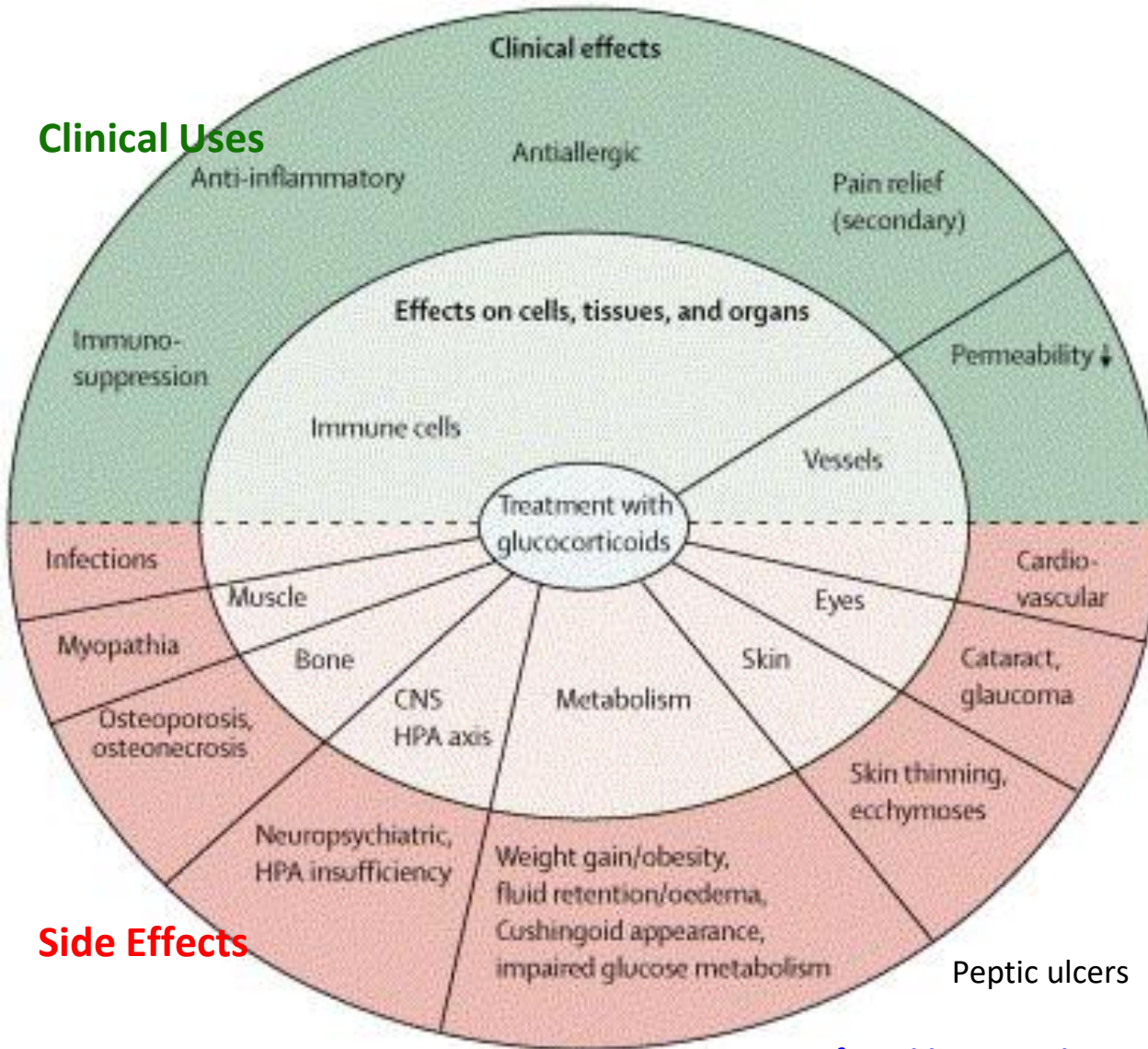


Caution

Appropriate use: Prior to use, the dose and duration of treatment should be based on the risk versus benefit for each individual patient. In general, **use the smallest effective dose** for the **shortest duration** of time to minimize adverse events. A gradual **tapering of dose** may be required prior to discontinuing therapy.

Benefit vs Risk

Clinical Uses

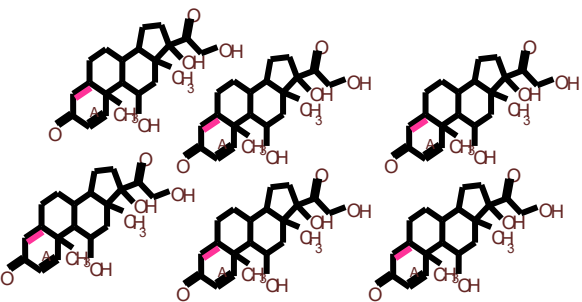


- Monitor patients for hyperglycemia, glycosuria, sodium retention, hypertension, hypokalemia, peptic ulcer, osteoporosis, and hidden infections
- Keep dose as low as possible
- Use short time periods (< 2 weeks)
- Decrease dose slowly if used for long periods

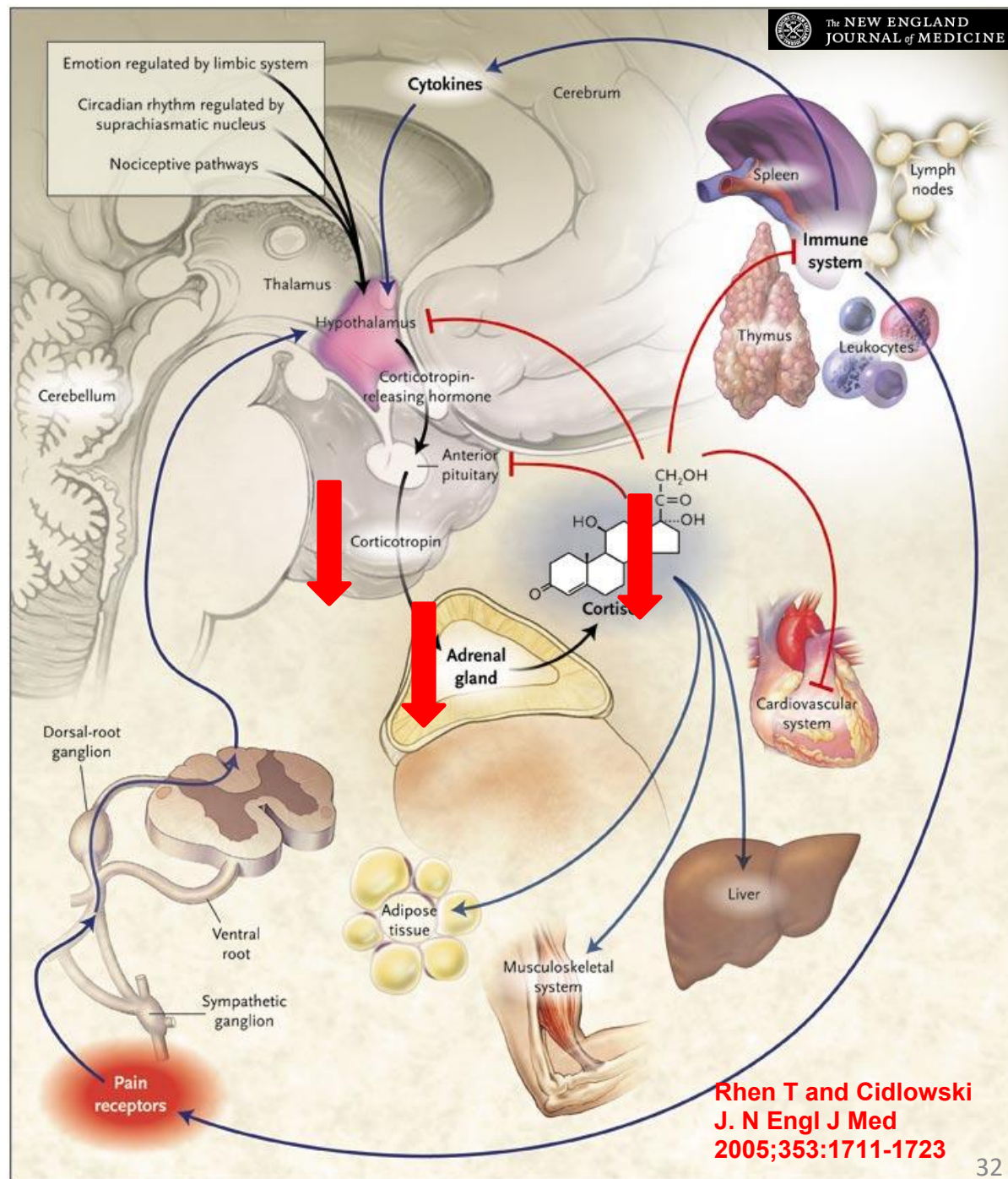
focosi.immunosig.org/immunomodulatory.html

Adrenal Suppression

Long term use results in feedback suppression of endogenous cortisol production

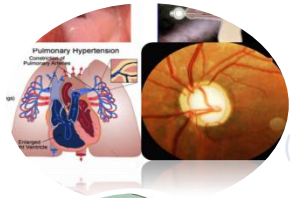


- ❖ Less likely to respond to stress from trauma or surgery
- ❖ Taper dose slowly to allow time (2-12 M) for endogenous pathways to reregulate
- ❖ Stopping dose too abruptly can lead to reappearance of symptoms with increased intensity



Rhen T and Cidlowski
J. N Engl J Med
2005;353:1711-1723

Anti-inflammatory Pharmacology



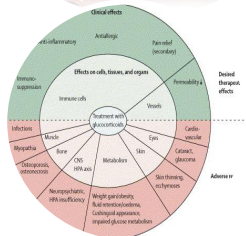
Eicosanoids

(Prostaglandins, thromboxanes, and Leukotrienes)

Biosynthesis

Cellular and tissue effects

Clinically relevant pharmacology



Glucocorticoids

(steroidal anti-inflammatory drugs)

Normal Function/intracellular mechanism

Anti-inflammatory effects

Types of synthetic glucocorticoids

Clinical pharmacology (uses and toxicity)



NSAIDs

(Non-steroidal anti-inflammatory drugs)

Role in blocking eicosanoids-mediated effects

Cost vs. Benefit of the three general classes

NSAID Choice



DMARDs

(Disease modifying anti-rheumatic drugs)

How do we stop disease progression with the least amount of harm?

Drugs used for gout Acute and chronic treatment



Eicosanoids

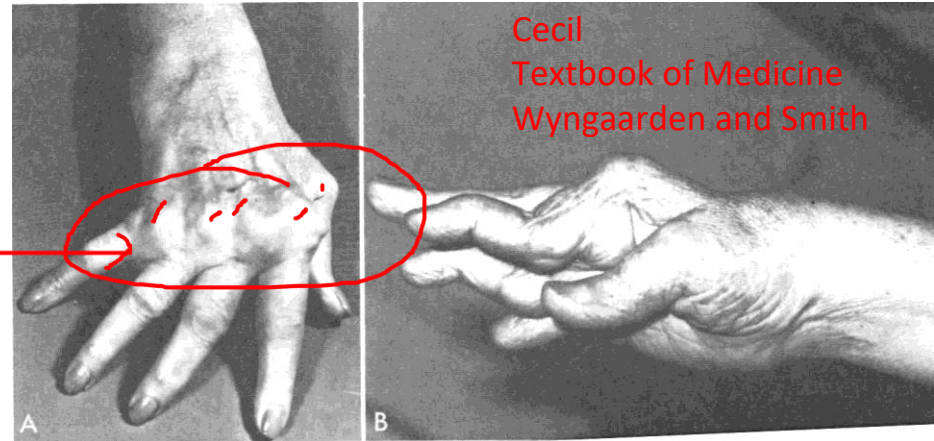
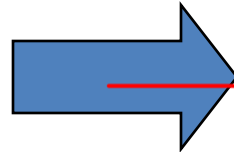
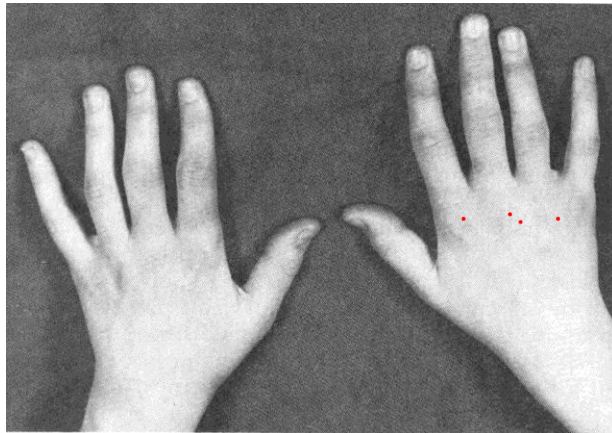
Glucocorticoids

NSAIDs

DMARDs

Anti-Gout

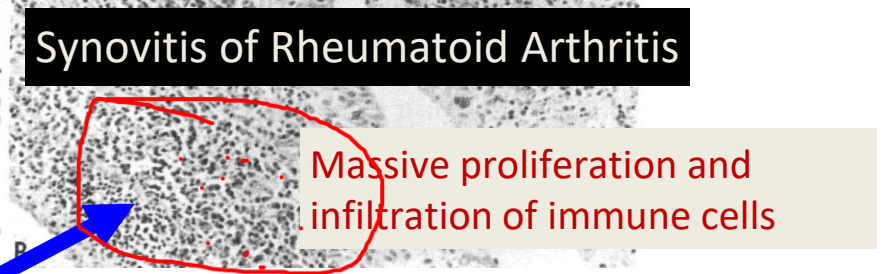
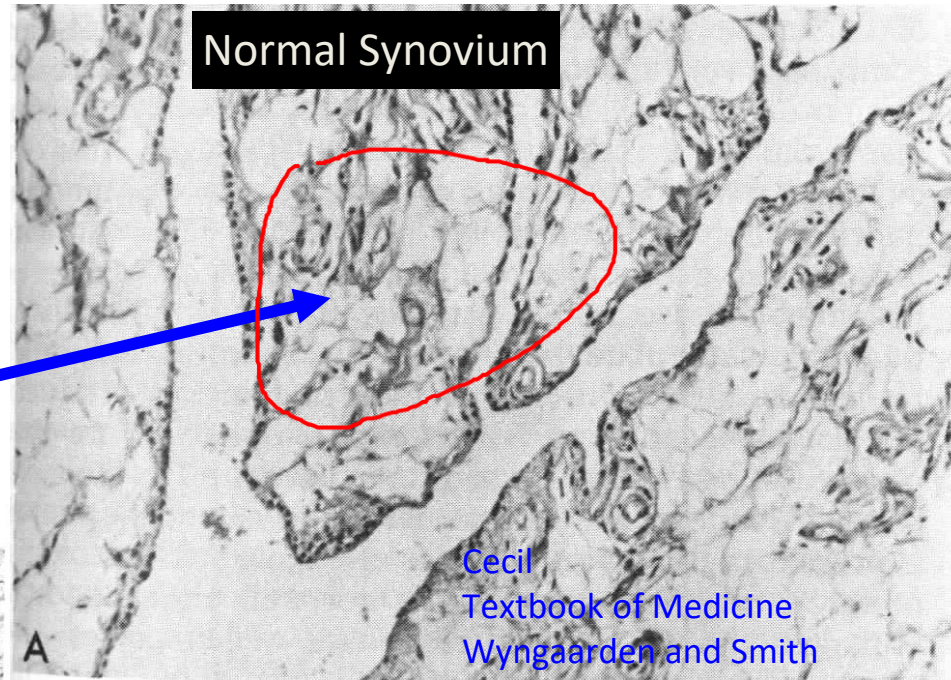
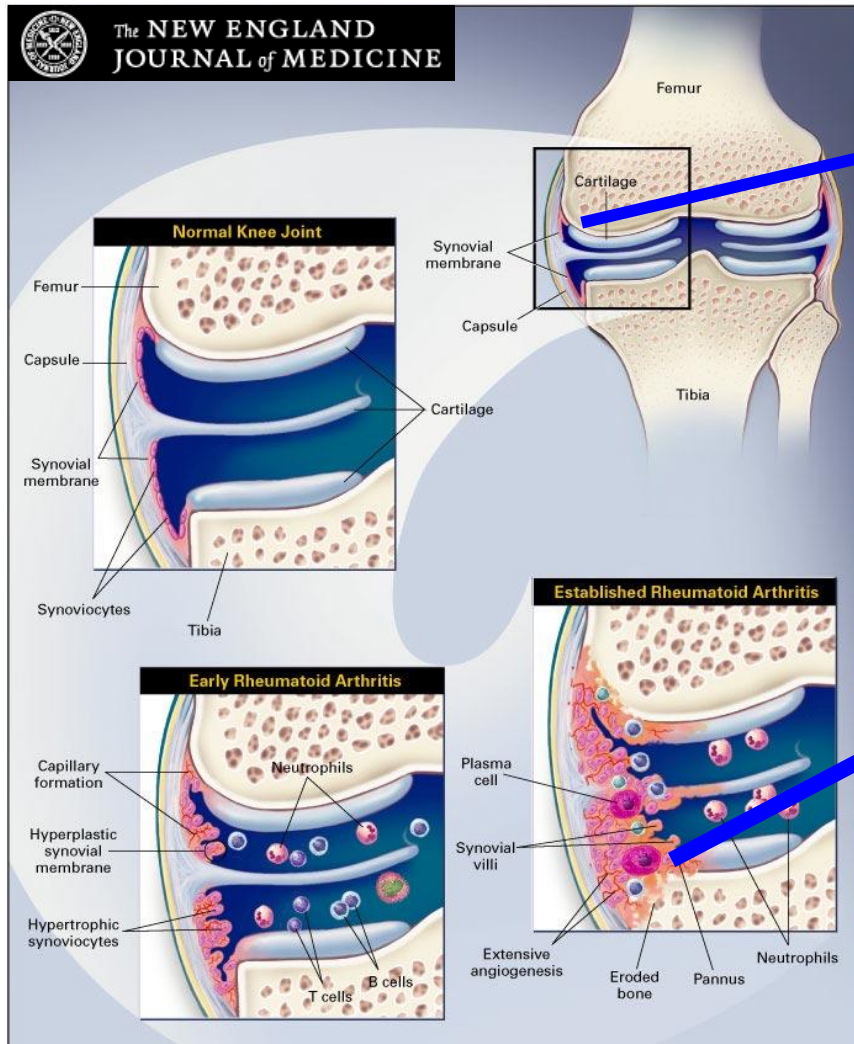
Rheumatoid Arthritis



- Rheumatoid arthritis
 - Immunologic disease
 - Reduces quality and quantity of life

Persistent morning stiffness
Swelling (symmetric and/or multiple joints)
Subcutaneous nodules
Serum Rheumatoid factor
Joint Erosion

Pathogenesis of Rheumatoid Arthritis



Thickened synovium due to hyperplastic, hypertrophic synoviocytes and is infiltrated with lymphocytes, plasma cells, macrophages and multinucleated giant cells. Many growth factors involved: Interleukin-1, GM-CSF, TNF- α , interferons, PDGF, etc.

Therapeutic Management of Rheumatic Diseases

OMM (Palliative therapy only)

NSAIDS

Steroids

DMARDS

- Decrease pain
- Improve mobility
- Improve Function

- Stop disease progression (when treated early)

Opioids

Tricyclic antidepressants

DMARDS



NSAIDs and Rheumatoid Arthritis

- Helpful in the first few weeks of patients **symptoms**
- Do not slow progression
- Should be used in combination with DMARDs
 - GI bleeding can be dangerous
 - COX2 specific can decrease this side effect
 - Proton pump inhibitors may help



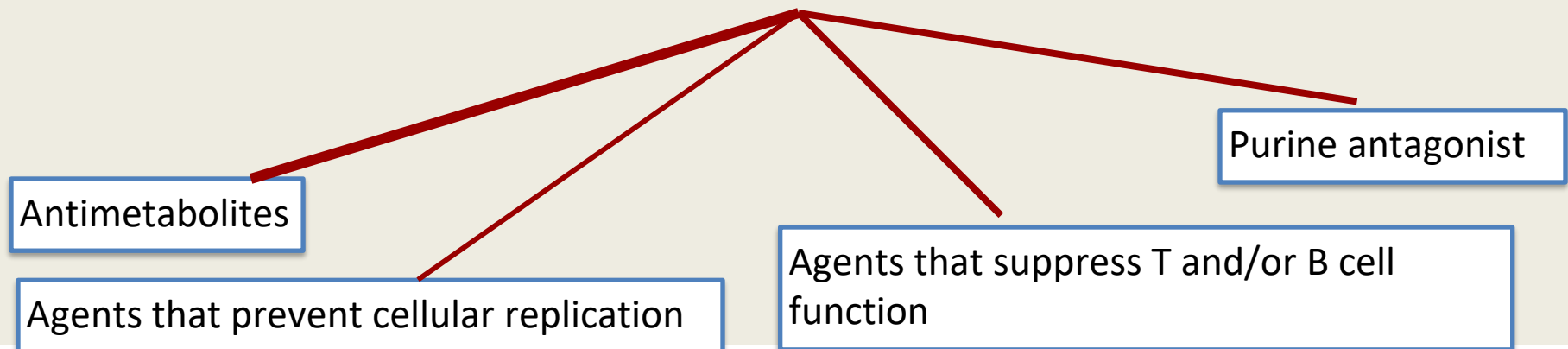
Glucocorticoid Drugs

- Slows the appearance of new bone erosion. Can be used initially while waiting for effects of DMARD therapy—"Bridge therapy"
- Adverse effects of prolonged use:
 - Extensive with prolonged use
 - Never exceed 10 mg daily

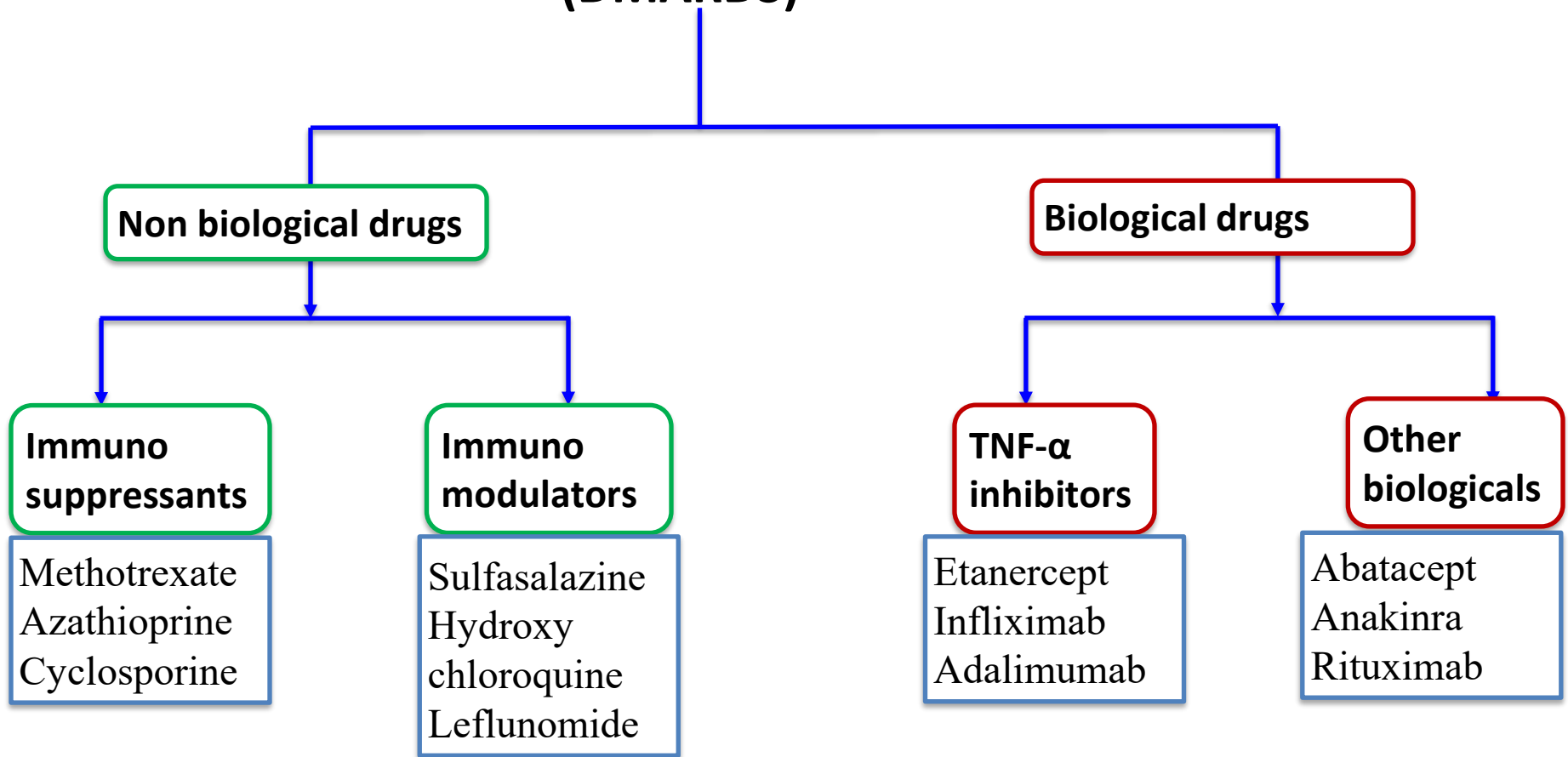
DMARDS

Immunosuppressive drugs:

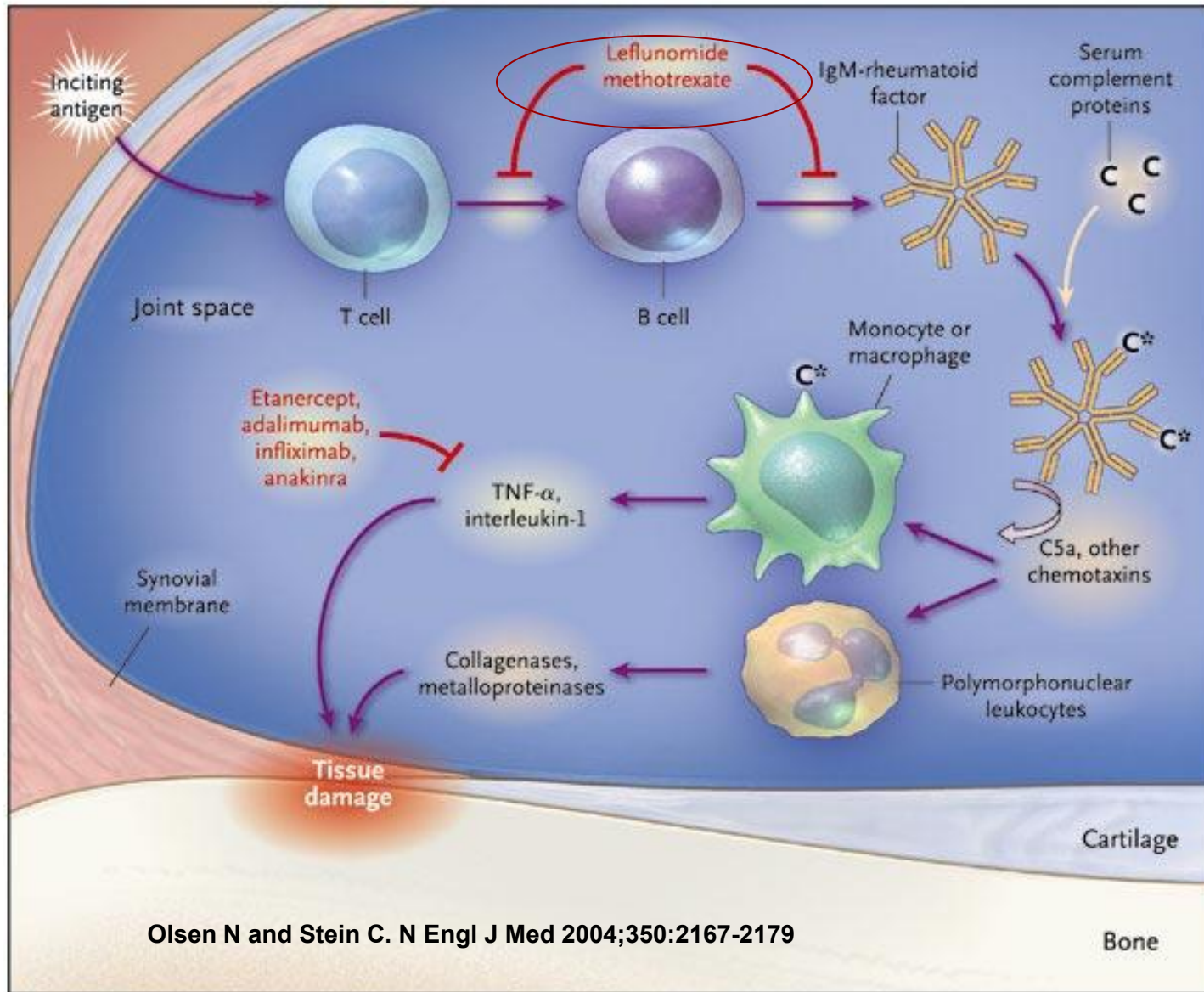
- Inhibit function and division of immune cells (or rapidly dividing cell populations)
- Started soon after RA diagnosis
- May take several months to show effects (slow-acting)
- Efficacy of methotrexate, sulfasalazine, IM gold, and penicillamine is similar
- Antimalarials are less effective



Disease modifying anti-Rheumatoid drugs (DMARDs)



Cellular/molecular targets



Cancer Chemo Drugs Used to Treat RA

Drug Type	Name	MOA	Kinetics	Toxicity
Anti-metabolite	*Methotrexate	Folic acid analogue inhibits (low dose) ***AICAR transformylase and TS and (High dose) dihydrofolate reductase, which is required for new DNA synthesis	Oral or intrathecal	inhibits all fast growing cells (gut epithelium {ulcers}, bone marrow blood progenitor cells, teratogen—Cat X), hepatotoxicity (dose dependent). GI and hepatic toxicity reduced with leucovorin
	**Azathioprine	Metabolite 6-thioguanine inhibits inosinic acid synthesis and inhibits B and T cell function, immunoglobulin production, and IL-2 secretion	oral	Bone marrow tox., oncogenesis, teratogenesis—Cat D, increased infection risk
DNA cross-linkers	Cyclo-phosphamide	Anti-neoplastic drugs used at low doses. Mechanisms cross-links DNA in rapidly dividing cells. Suppresses B and T cell function by 30-40%	oral	Bone marrow suppression, immunosuppression (Cat X)

*DMARD of first choice. Most often selected as initial therapy--Lowers mortality rate.

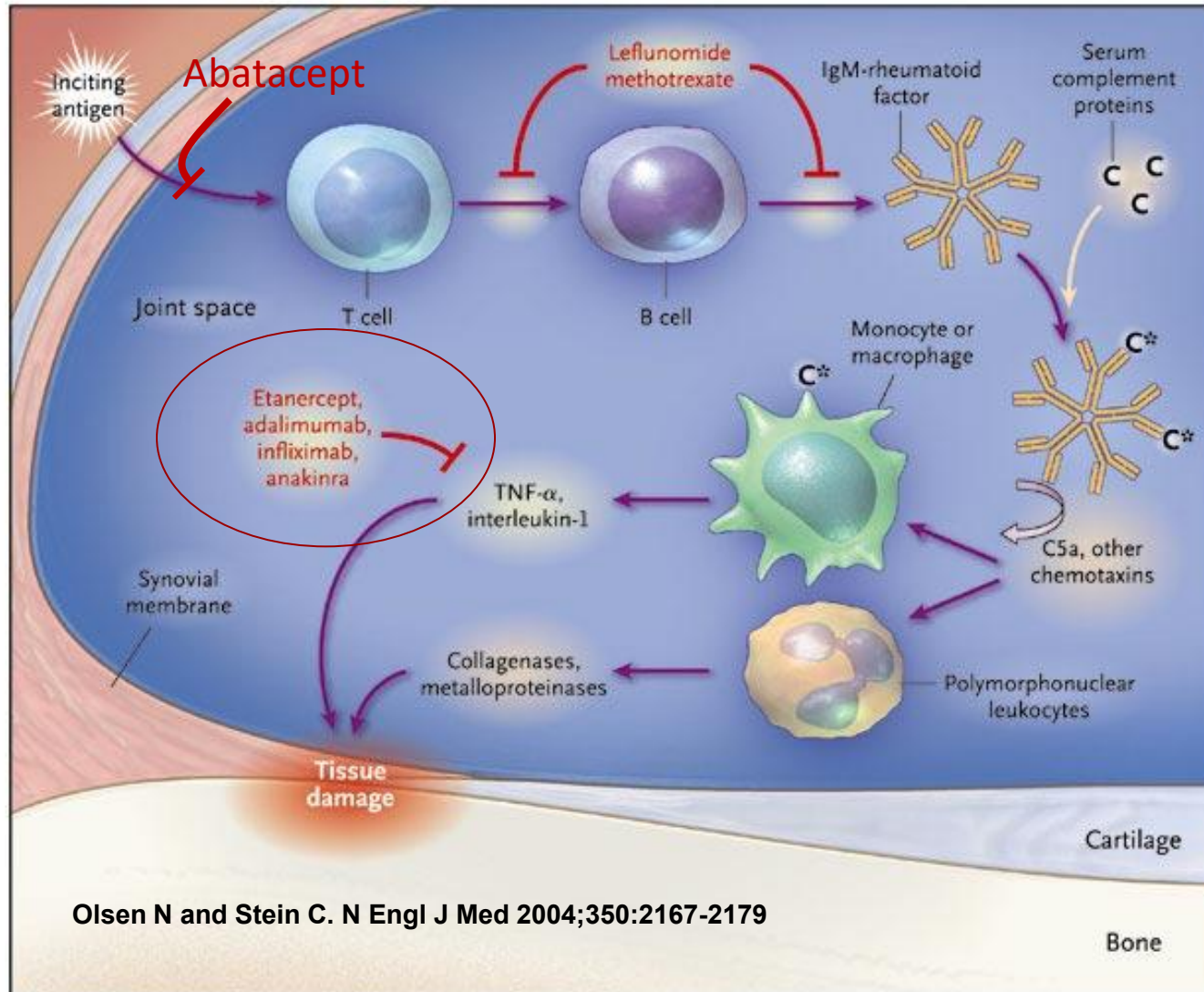
** Also used for many other inflammatory conditions (psoriasis, reactive arthritis, polymyositis, SLE, Behcet's disease)

** aminoimidazole carboxamide ribonucleotide (AICAR), thymidylate synthase (TS)

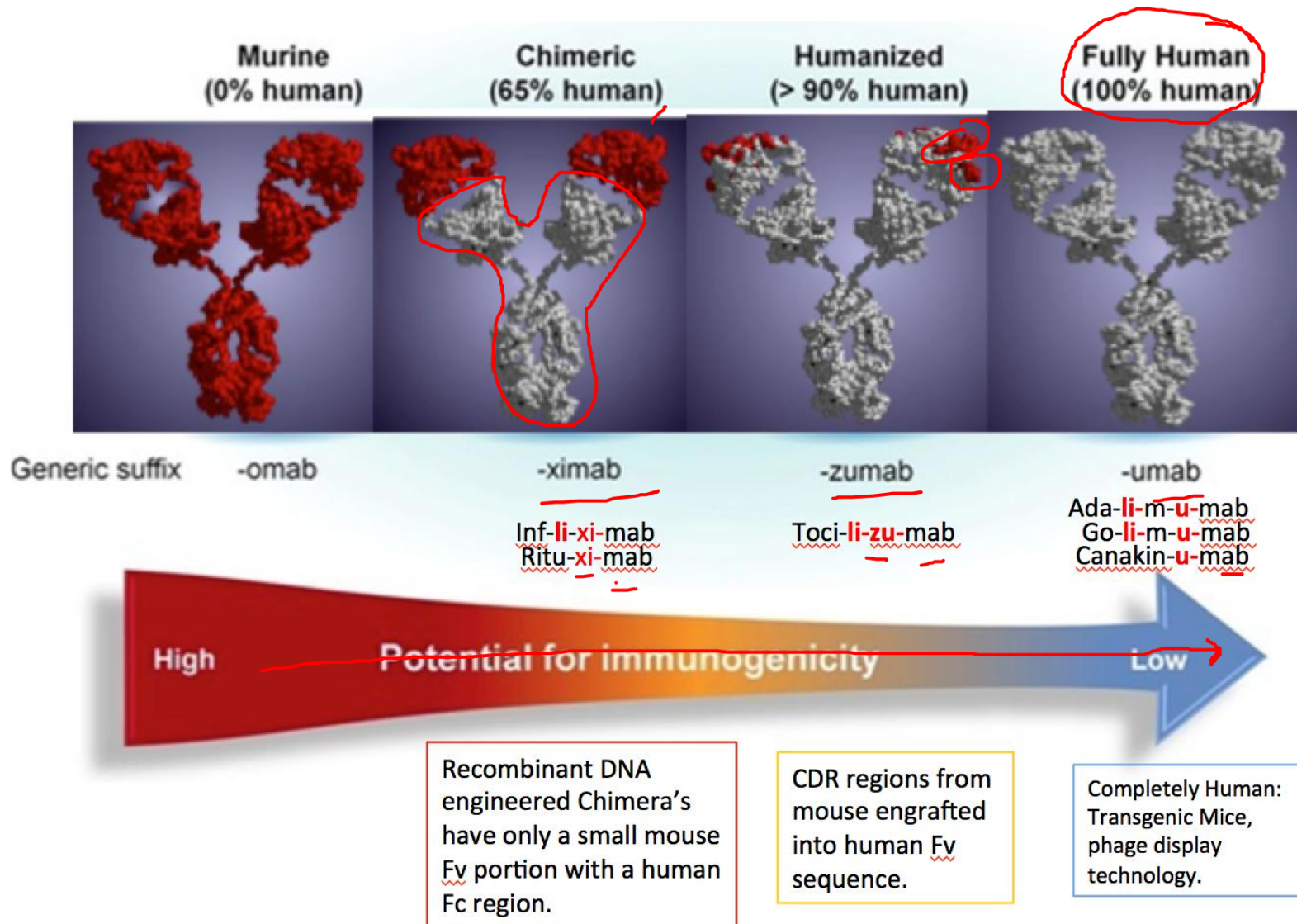
Drugs that ↓ Lymphocyte and Macrophage function

General Class	Name	MOA	Kinetics	Toxicity
	*Leflunomide * Efficacy similar to MTX	Metabolite inhibits dihydro orate dehydrogenase inhibition de novo ribonucleotide synthesis, stops T cell proliferation and B cell function	Oral, active metabolite (A77-1726)	Adverse effects: diarrhea (25%), rash and elevated liver enzymes and even hepatotoxicity (0.28%) . Mild alopecia, weight gain, and increased blood pressure Fetal death, teratogen—Cat. X
Anti-malarial	Hydroxy chloroquine (less effective)	unclear (antimalarial), inhibits leukocyte function	Oral (takes months to show effect)	Retinopathy, GI distress, ototoxicity, headache, myopathy and neuropathy. Cat C
Gold Salts Less favored toxicity	Auro-thiogluucose	inhibits macrophage function	IV/IM weekly	dermatitis (15-20%), cyto-penias (1-10%), proteinuria (8-10%), GI disturbances common (diarrhea),
	Auranofin		Oral	
	Sulfasalazine	inhibition of B cell function, main use =ulcerative colitis, inflammatory bowel disease	Metabolized to sulfapyridine (RA) and 5-aminosalicylic acid (IBD) Oral	30% patient discontinue use-- Photosensitivity, GI (nausea), cytopenia, skin rash, drug induced lupus
	Mycophenolate Mofetil	Inhibits inosine monophosphate dehydrogenase which suppresses B and T cell function. Interferes with leukocyte adhesion	oral	Dyspepsia and abdominal pain, hepatotoxicity. Bone marrow suppression, increased infections and rarely associated with malignancy
	Cyclosporine	Covered in immuno pharm lecture		

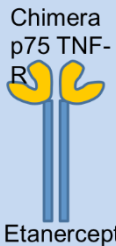
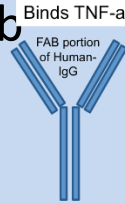
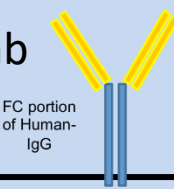
Biologic DMARDs



Side Note: Humanized Antibodies



Drugs that inhibit TNF- α function

Name	MOA	Kinetics	Toxicity
Etanercept 	TNF- α receptors linked to IgG1	SC 2x week	1) Serious infection risk: $\uparrow$ in macrophage-dependent infections (like TB, fungal, viral and opportunistic infections). Assess for latent or active TB. 2) Associated with HBV reactivation. 3) Rare cases of leukopenia and vasculitis 4) Drug induced lupus (?) 5) Exacerbate HF 6) Immune response to drugs 7) $\uparrow$ risk of skin cancer (screen). Potential increased risk of non-Hodgkin's lymphoma?
Adalimumab Golimumab 	Fully Human monoclonal against TNF- α	IV weekly	
Infliximab 	Mouse FAB against TNF- α , FC human	IV monthly	
Certolizumab,	Human FAB against TNF- α linked to Polyethylene glycol	SC 2x week	



Eicosanoids

PEG

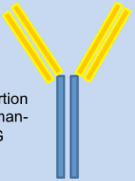
Glucocorticoids

NSAIDs

DMARDS

Anti-Gout

Other Cytokine Function Inhibitors

Name	MOA	Kinetics	Toxicity
Tocilizumab 	Humanized AB. Binds to IL-6 receptors preventing activation of IL-6 cell signaling (prevents T-Cell activation and other inflammatory processes involved in RA)	IV q4wk IL-6 suppresses several CYPs -- inhibiting will ↑ CYP activity = drug interactions.	1) Serious infection risk: ↑ in macrophage-dependent infections (like TB, fungal, viral and opportunistic infections). Assess for latent or active TB. 2) Neutropenia (rare) 3) Altered lipid profile
Anakinra	Recombinant IL-1 RA -- antagonizes IL-1 receptors. (more for other inflammatory diseases, like gout & CAPs, than RA)	SC daily	1) Increase in serious infection risk 2) Adverse effects: injection site irritation (50-80%) 3) Transient neutropenia (monitor)
Riloncept	Ligand binding domain of IL-1 thus neutralizing IL-1 by preventing binding to cell receptor (used for CAPs & Gout)	SC q4wk	
Canakinumab	Human IgG MAB to IL-1 β — prevents binding of IL-1 to receptor	SC qwk	

Eicosanoids

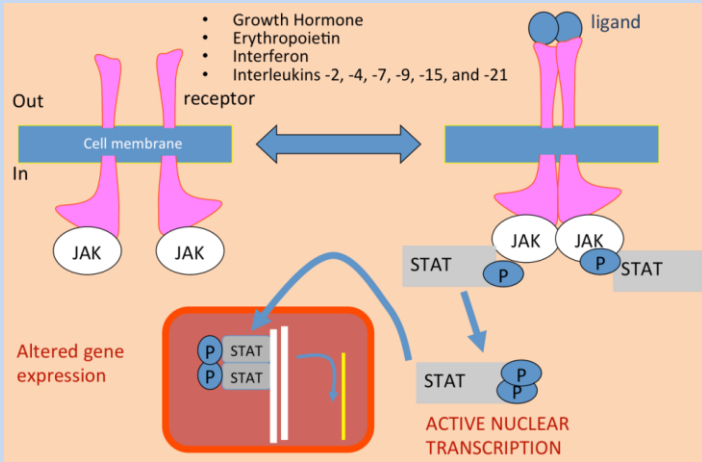
Glucocorticoids

NSAIDS

DMARDS

Anti-Gout

T or B-lymphocyte Inhibitors

Name	MOA	Kinetics	Toxicity
Rituximab	Chimeric MAB binds CD20 on B cells—compliment activation and B cell depletion	IV infusion, every 2 weeks	1) Serious infection risk: ↑ in macrophage-dependent infections (like TB, fungal, viral and opportunistic infections). Assess for latent or active TB. 2) Altered lipid panels 3) Tofacitinib ↑ risk of cancer (screen) 4) Rituximab-rash (30%) 5) Rituximab reactivation of HBV
Abatacept	Chimera: extracellular domain of human CTLA-4 linked to modified human IgG. Inhibits stimulation of T-Cells by blocking CD28-CD80/86	IV infusion every 2-4 weeks	
Tofacitinib	Synthetic small molecule that inhibits JAK/STAT signaling which plays major role in pathogenesis of RA  • Growth Hormone • Erythropoietin • Interferon • Interleukins -2, -4, -7, -9, -15, and -21 Out receptor Cell membrane In JAK JAK ligand JAK JAK STAT P STAT STAT P Altered gene expression ACTIVE NUCLEAR TRANSCRIPTION	Oral q2xdaily	

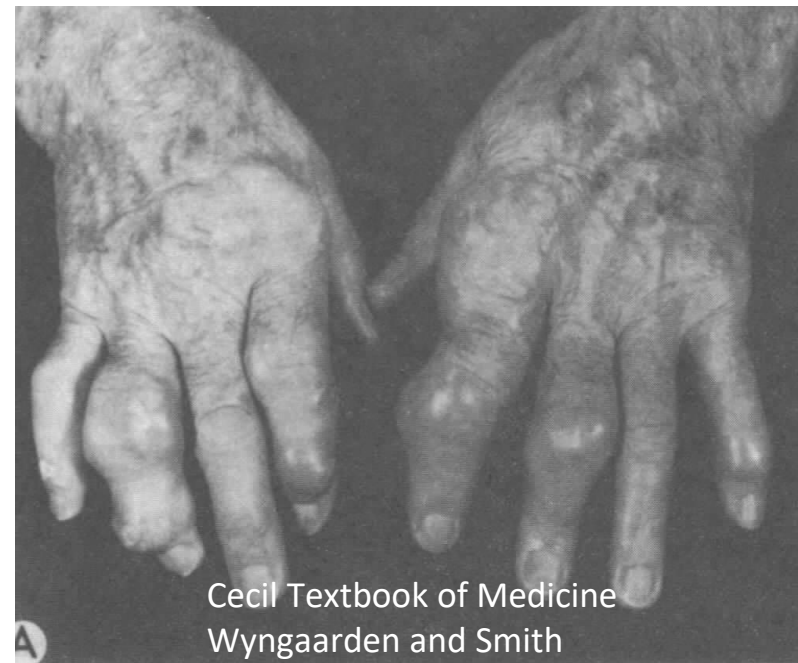
DMARDS Prolong Life with side effects

- Start therapy immediately after confirming diagnosis
- Monitor efficacy of therapy: American College of Rheumatology (ACR) criteria
- Combinations therapy may be more efficacious
 - Use combinations if MTX alone fails to improve disease activity by greater than 20%
- Monitor liver function
 - Serum albumin and aminotransferase levels every 4-8 weeks
- Obtain complete blood counts
- Measure serum creatinine
- Avoid pregnancy –most
 - RA symptoms fortunately abate during pregnancy with postpartum exacerbation—if you have to use something low dose glucocorticoids, hydroxychloroquine or sulfasalazine.
- Administer Influenza and pneumococcal vaccination



Gout

- Familial metabolic disease
 - Associated with hyperuricemia
 - recurrent episodes of acute arthritis due to monosodium urate deposits in joints and cartilage
 - high serum uric acid levels
 - an end product of purine metabolism



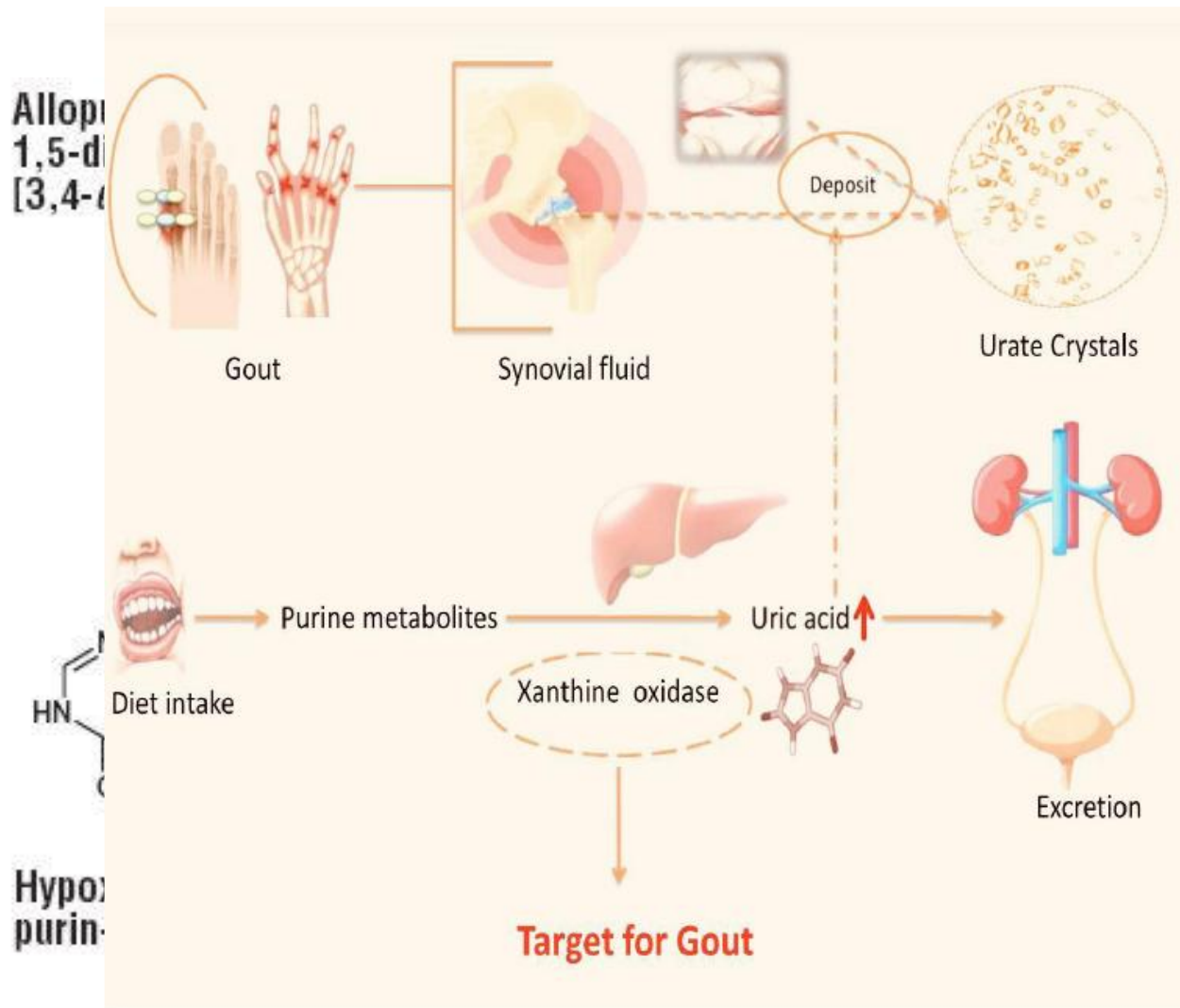
Glucocorticoids

NSAIDS

DMARDS

Anti-Gout

Purine Synthesis



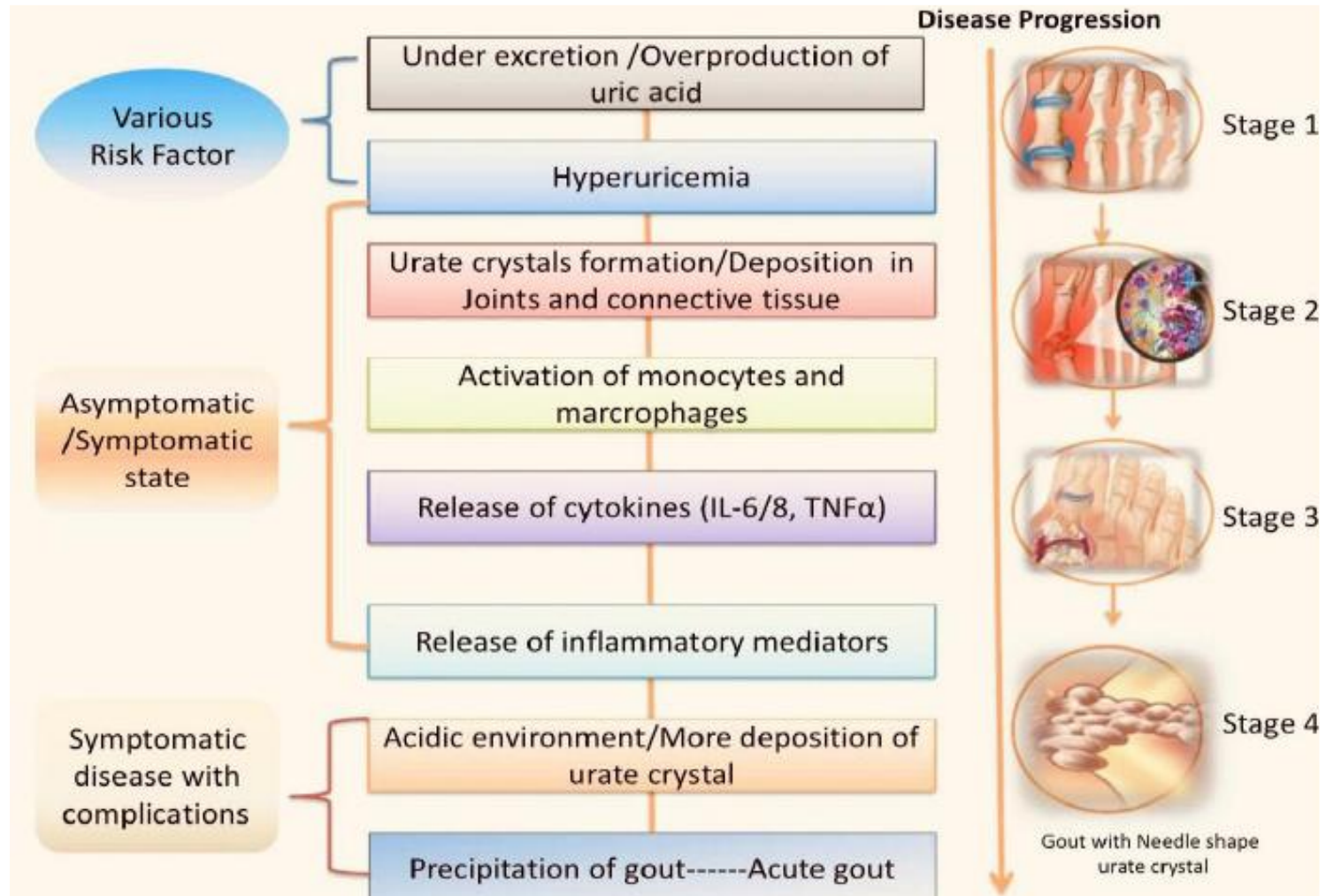
Glucocorticoids

NSAIDS

DMARDS

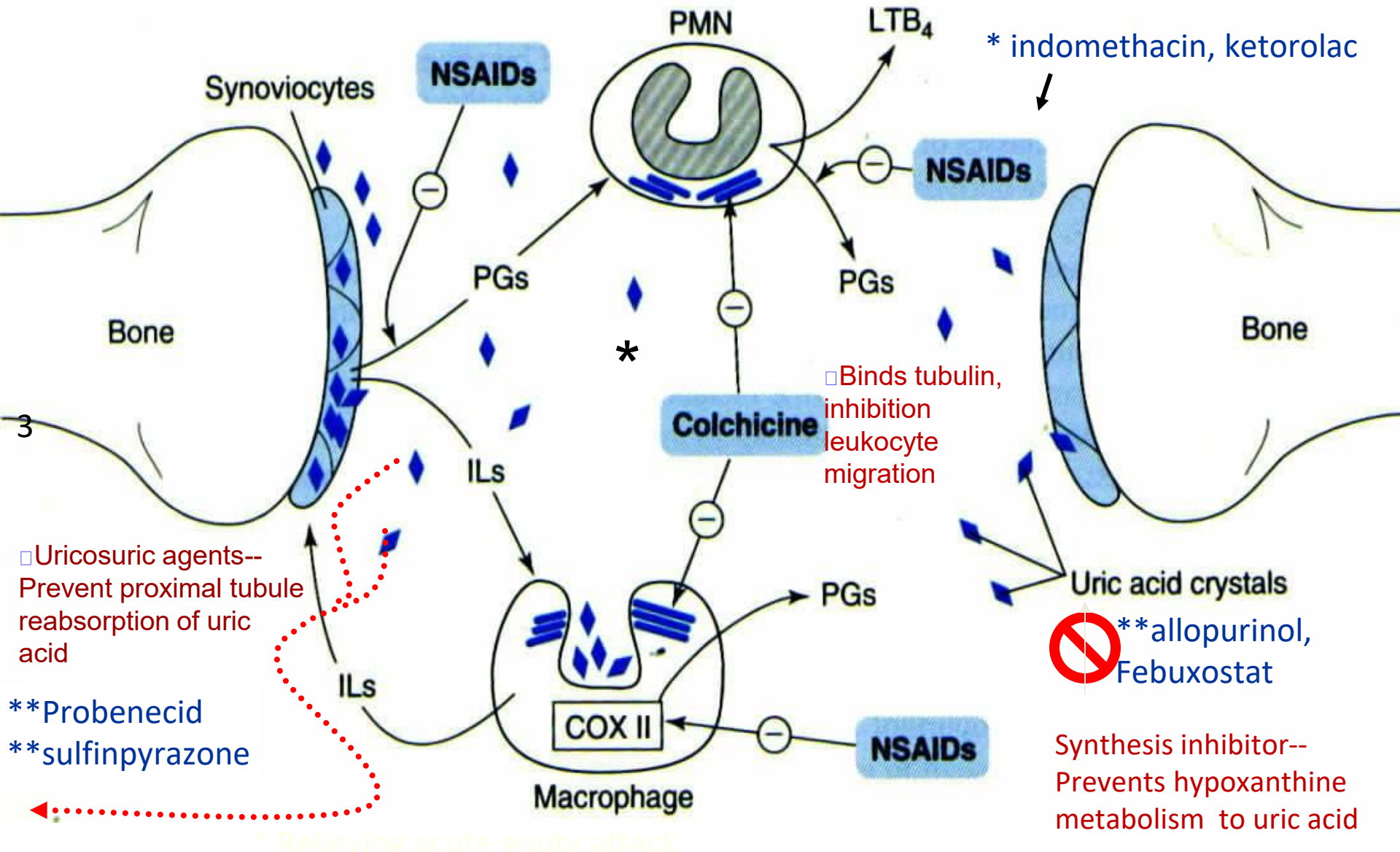
Anti-Gout

Pathogenesis and progression of gout



Gout Therapies

Examination & Board Review
Pharmacology: Katzung, Appleton and Lange



Eicosanoids

Glucocorticoids

NSAIDS

DMARDS

Anti-Gout

Drugs used to treat Gout

Name	Use for Gout	MOA	Kin.	Toxicity
Colchicine	Prophylaxis for recurrent episodes. Treat acute gouty attacks	inhibits microtubule polymerization by binding to tubulin inhibition of leukocyte migration	oral	10%: Gastrointestinal: Gastrointestinal disorders including abdominal pain, cramping, nausea, vomiting (up to 26%), diarrhea (up to 23%) (see UpToDate) rarely causes hair loss, bone marrow depression, peripheral neuritis and myopathy
Indomethacin	Treat Acute gouty attacks	NSAID		see NSAIDs
Anakinra Canakinumab, rilonacept	If refractory to colchicine and NSAIDs	IL-1 inhibitors	parenteral	See before

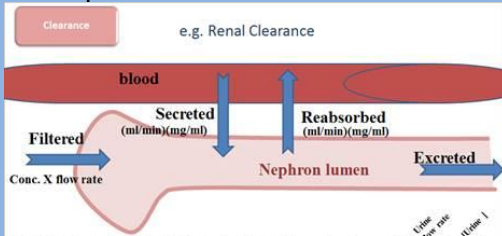
Glucocorticoids

NSAIDS

DMARDS

Anti-Gout

Drugs used to treat Gout

Name	Use for Gout	MOA	Kin.	Toxicity
Probenecid	several acute attacks of gouty arthritis, evidence of tophi, or extremely high plasma levels	↓ the bodies pool of urate Both are weak acids and can compete with uric acid (a weak acid) at proximal renal tubule weak acid active transport sites preventing reabsorption of uric acid: Maintain high urine volume to avoid precipitation of crystals	oral	GI irritation Probenecid can cause allergic dermatitis and nephrotic syndrome
sulfinpyrazone				
				
Allopurinol Febuxostat	chronic tophaceous gout w/ elevated uric acid levels or in combo with	inhibits xanthine oxidase, prevents urate formation. From dietary purines (allopurinol is a purine analog while febuxostat is not)		Precipitates acute attacks of gouty arthritis (Colchicine or NSAIDS she used prophylactically) GI intolerance, Bone marrow depression (leukocytosis—3%), Hepatotoxic (68%), Skin rashes (92%) Febuxostat is better tolerated than allopurinol
Pegloticase	Refractory chronic gout	Recombinant urate oxidase linked to methoxy polyethylene glycol, when injected converts uric acid into allantoin	IV every 2 weeks	Infusion reaction and gout flare up, Other nephrolithiasis, arthralgia, muscle spasm, headache, anemia and nausea

Glucocorticoids

NSAIDS

DMARDS

Anti-Gout

Gout treatment

Acute Gout Flare Management

- NSAIDS (Ibuprofen, naproxen)- Reduce inflammation and pain
- Colchicine: Alleviates inflammation, especially started early in the flare
- Corticosteroids: If NSAIDs or colchicine are ineffective

Chronic Gout Management (prevent recurrent flares)

- Lifestyle modifications:

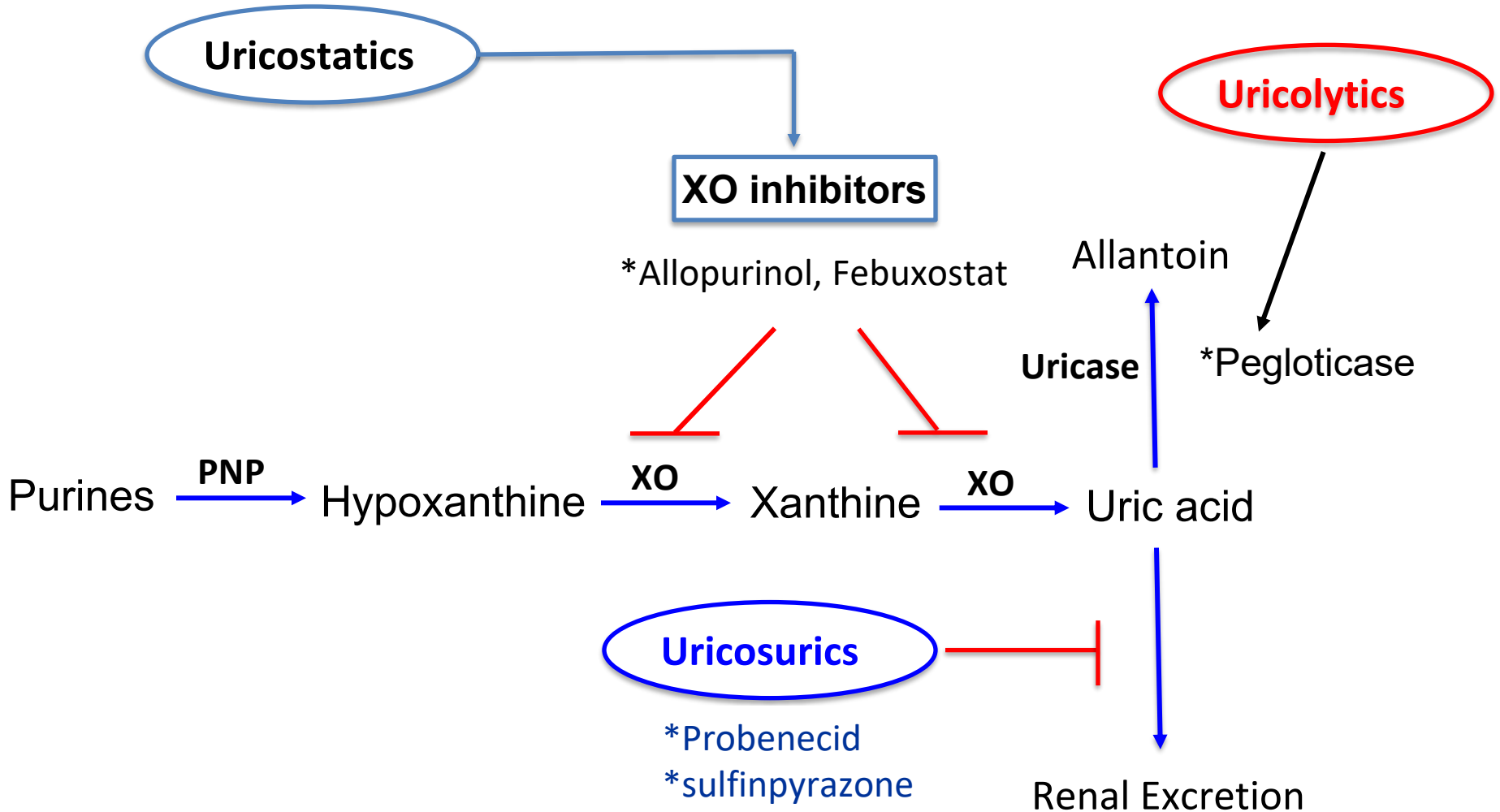
- Diet: Limit purine rich foods (red meat, seafood) and alcohol (beer)
- Hydration: maintain good hydration to aid uric acid excretion
- Weight management: reduce weight if obese

Pharmacologic Uric Acid Lowering Therapy

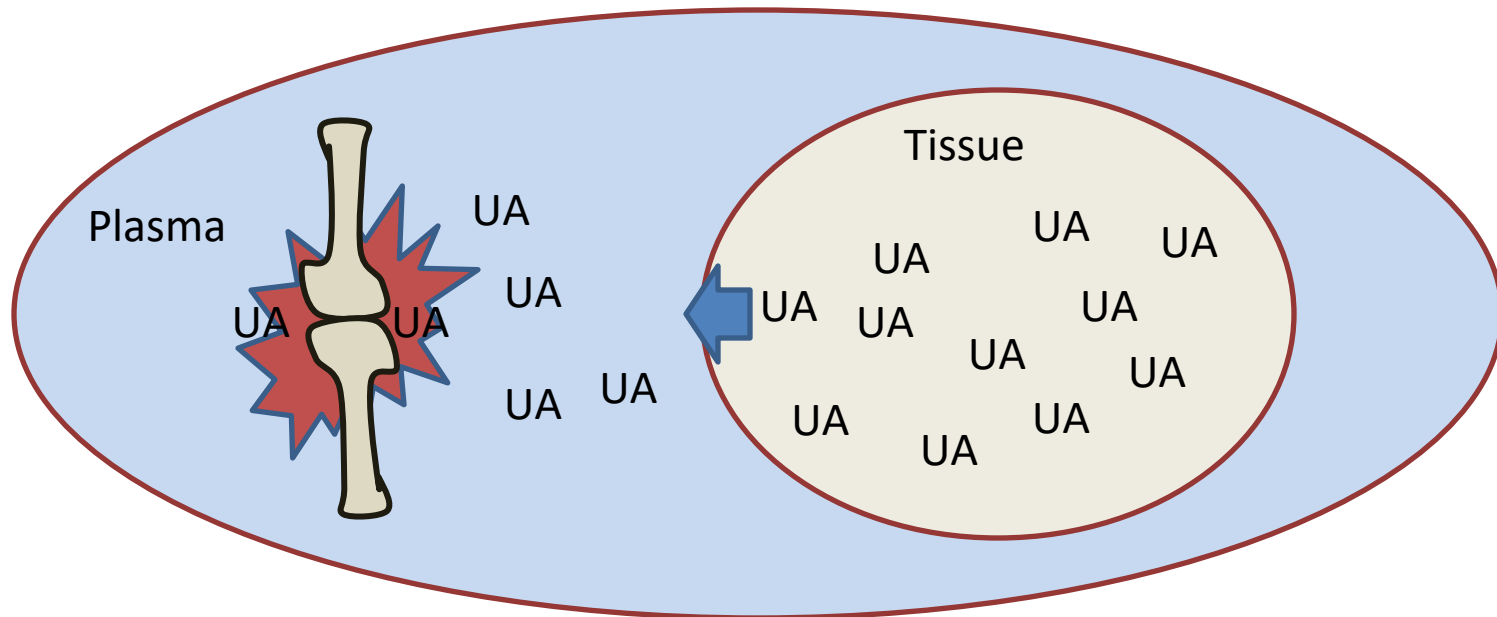
- Allopurinol or Febuxostat: Xanthine Oxidase inhibitors to reduce uric acid production
- Probenecid: Enhances renal excretion of uric acid (unable to take XOI)
- Pegloticase: Enzyme that breaks down uric acid: in refractory cases



Gout treatment



Gout + Allopurinol (acute)



Acute Allopurinol or Uricosurics can precipitate an acute gouty flare up. Use colchicine or NSAID to avoid flare up

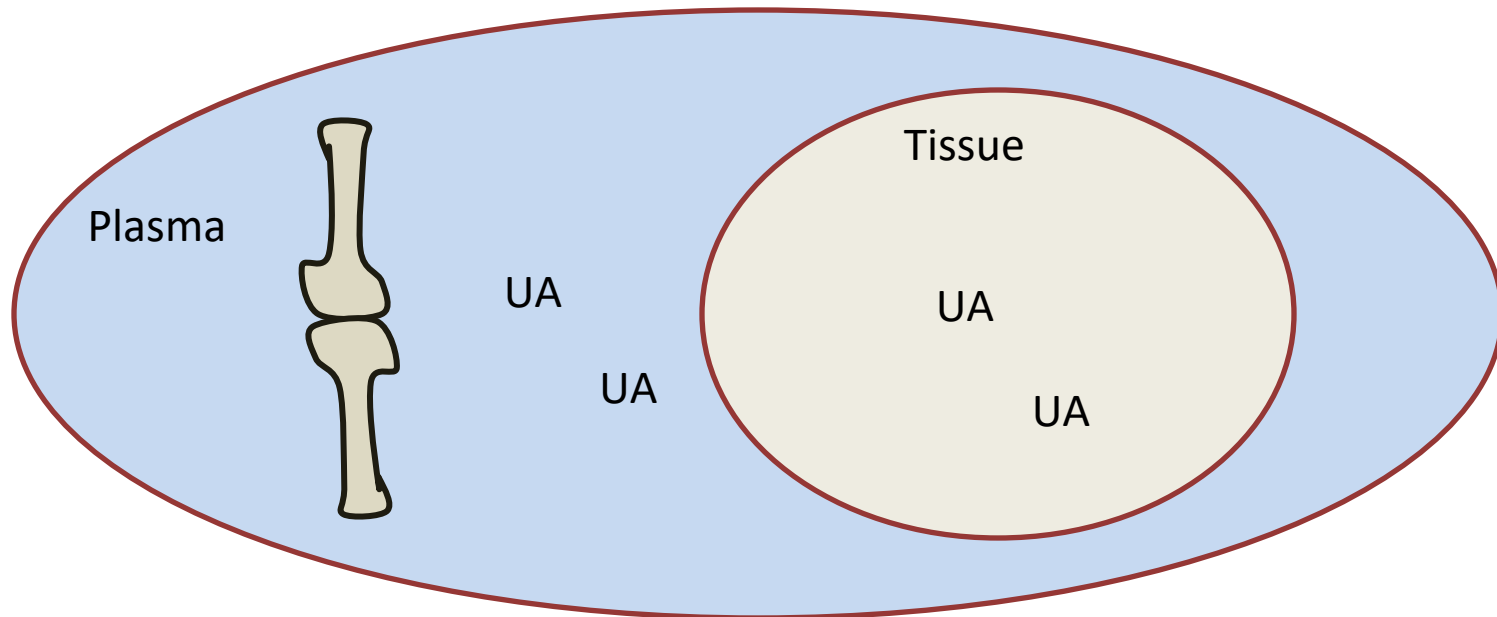
Glucocorticoids

NSAIDS

DMARDS

Anti-Gout

Gout + Allopurinol (chronic)



Several weeks to decrease tissue levels
fluctuations can last for months



Study Goals

NSAIDS

Be able to relate the effects of NSAIDS on lipxygenase, cyclooxygenase (COX1 and COX 2) and the consequences of the products produced with clinical features including use and toxicity

Distinguish the clinical importance of selective COX2 vs. COX1 inhibition or nonselective inhibition.

Relate the use aspirin, NSAIDs and coxibs to the clinical toxicity

Categorize the major NSAIDS and coxibs by name

Be familiar with NSAIDS with long half lives

DMARDs

Compare and contrast the various DMARDs, mechanism of action, use, and toxicities and relate these parameters to decision regarding appropriate management choice.

Anti-Gout

Compare and contrast the various anti-gout medications, mechanism of action use and toxicity and relate these parameters to decision regarding appropriate management based on disease severity.

Drug List

•Glucocorticoids

Cortisone

Betamethasone

Dexamethasone

Hydrocortisone

Prednisolone

Triamcinolone

Anti-inflammatory agents

•Cox-2 Selective inhibitors

–Celecoxib *Celebrex*

–Meloxicam *Mobic*

–Valdecoxib *Bextra*

•Nonselective cox inhibitors

–Aspirin

–Diclofenac *Cataflam*

–Diflunisal *Dolobid*

–Etodolac *Lodine*

–Fenoprofen *Nalfon*

–Flurbiprofen *Ansaid*

–Ibuprofen *Motrin, Advil*

–Indomethacin *Indocin*

–Ketoprofen *Orudis*

–Ketorolac *Toradol*

– Meclofenamate &
mefenamic acid *Ponstel*

– Nabumetone *Relafen*

– Naproxen *Naprosyn, Aleve*

– Oxaprozin *Daypro*

– Piroxicam *Felden*

– Sulindac *Clinoril*

– Tolmetin *Tolectin*

Drug List cont'd

DMARDS

•Synthetic

- Methotrexate
- Leflunomide
- Cyclosporine
- Hydroxychloroquine
- Gold
- Penicillamine
- Sulfasalazine
- Mycophenolate mofetil

•Biologic

- Infliximab
- Adalimumab
- Etanercept
- Rituximab
- Abatacept
- Golimumab
- Certolizumab
- Tocilizumab

- Anakinra

Anti-Gout Agents

- Colchicine
- Probenecid
- Sulfinpyrazone
- Allopurinol
- Febuxostat
- Pegloticase